Renewable and Sustainable Energy Reviews 145 (2021) 111111

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: http://www.elsevier.com/locate/rser

# Power system stability issues, classifications and research prospects in the context of high-penetration of renewables and power electronics

Jan Shair [a], Haozhi Li [a], Jiabing Hu [b], Xiaorong Xie [a,*]

[a] State Key Laboratory of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing, 100080, China
[b] State Key Laboratory of Advanced Electromagnetic Engineering and Technology, School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China

ARTICLE INFO

*Keywords:*
Classification
Power electronics
Power system stability
Stability issues
Renewable power generators

ABSTRACT

This paper concerns with the emerging power system stability issues, classification, and research prospects under a high share of renewables and power electronics. The decades-old traditional power system is undergoing a fast transition with two most prominent features: 1) high-penetration of renewable power generators, utilizing intermittent renewable sources such as wind and solar, and 2) high-penetration of power electronic devices in the generation e.g., wind turbine converters and solar power inverters, transmission e.g., flexible ac or dc transmission system converters, and distribution/utilization systems e.g., electric vehicle and microgrid. The development of modern power systems with dual high-penetrations, i.e., high-penetrations of renewables and power electronic devices, influences the power system dynamics significantly and causes new stability issues. This paper first overviews equipment-level features and system-level stability challenges introduced under the dual high-penetration scenario of the modern power system. Next, the impacts of emerging stability challenges on various aspects of the classical stability issues and classifications are highlighted. Under this context, this paper discusses the validity and limitations of the existing classical and extended power system stability classifications proposed by different IEEE/Cigre Working Groups. Furthermore, a new power system stability classification framework is proposed, which not only maintains the inherent logic of the classical classification but also provides wide coverage and future adaptability of the emerging stability issues. Finally, various classification-oriented research prospects in the power system stability domain are highlighted.

## 1. Introduction

The increased environmental protection awareness and related government policies to meet ever-increasing energy demand has stimulated a paradigm shift from conventional turbogenerators (TGs) to renewables power generators, such as wind and solar. In the past nine years (2010–2019), the total installed capacity of renewable power in the world has increased from 1.2 TW to 2.5 TW [1]. In the same period, the installed capacity of renewable power in China increased from 0.2 TW to 0.7 TW [1]. It is estimated that, by 2030, the installed capacity of wind and solar photovoltaic power generation will reach around 3.8 TW in the world, and 2 TW in China [1]. The transition from conventional to renewable power generation at such a large scale is great because of the scientific and technological advancements in power electronic technologies [2]. The proportion of renewables and power electronics in the modern electric power system has increased tremendously and is expected to increase even more in the future [3].

With the advancements in power electronic technology in the past few decades, the power electronic converters have found applications in the generation, transmission, distribution, and utilization of electric power [4]. For instance, on the generation side, most of the installed wind and solar photovoltaic power generators employ power electronics in the form of wind turbine converters and photovoltaic inverters for grid connection, which hereinafter will be collectively called converter-interfaced-generators (CIGs). On the transmission side, the core modern power transmission technologies are also based on power electronic converters, such as flexible ac transmission system (FACTS) and high voltage dc transmission system (HVDC). For flexible and efficient transmission of electric power, the proportion of FACTS and HVDC in the power transmission system is increasing around the world. The distribution systems are also transforming due to emerging concepts with fast pace practical implementations, including distributed

* Corresponding author.
*E-mail addresses:* janshair@outlook.com (J. Shair), xiexr@tsinghua.edu.cn (X. Xie).

https://doi.org/10.1016/j.rser.2021.111111
Received 25 December 2020; Received in revised form 8 April 2021; Accepted 12 April 2021
Available online 19 April 2021
1364-0321/

generation (DG), dc distribution networks, and microgrids with power electronics being the core enabling technology [5]. In the utilization aspect, new loads such as variable frequency drives, power quality controllers, and electric vehicles also utilize power electronic converters [6]. According to a report published by the electric power research institute (EPRI), "the 21st century is the golden age for the application of power electronics technology", and its role will be as important as computing, communication, and information technology [7].

In a nutshell, with the technological advancements of renewable power generation and power electronics [10], the modern power system is going through a dual high-penetration development trend.

A power system with the *high-penetration* of intermittent renewable energy resources and the *high-penetration* of power electronic converters in the generation, transmission, distribution, and utilization is named as the **dual high-penetrated power system**; the term dual high-penetration will be used throughout this paper to represent high-penetration of renewables and power electronics.

The CIGs are essentially different from the traditional TGs, for example, low capacity, small size, low inertia, leading role of power converter control, and so on. Similarly, the dynamics of modern converter-dominated transmission and distribution networks are significantly different from conventional transmission and distribution networks. These differences have a profound impact on the overall dynamic stability of the power system, and thus, bring new stability problems and challenges to its safe operation [8].

The main contributions are as follows:

- First, this paper summarizes the device and system-level characteristics of the dual high-penetrated power systems, analyzes its impact on the classical power system stability, and discusses the emerging stability problems.
- Next, it provides a critical review of existing classical and extended definitions and power system stability classifications proposed in 2004 and 2020, respectively [9,10].
- Next, the paper proposes a new framework for the power system stability classification, which not only maintains the basic logic of the classical stability classification of 2004 but also provides a better framework for understanding the stability issues and future adaptability.
- Finally, it points out classification-oriented future research directions related to emerging stability issues in the context of dual high-penetrated power systems.

The authors anticipate that the paper will provide a great insight into the emerging stability issues and stability classifications and will promote future research in this domain.

The rest of the paper is organized into the following sections. Section 2 briefly overviews the device and system-level characteristics of the dual high-penetrated power system. Section 3 summarizes the influence of dual high-penetrations features on the classical stability issues and outlines emerging stability issues. Section 4 reviews the existing classical and extended power system stability classifications and highlights the limitations. Section 5 proposes a new stability classification and compares its features with the existing classification. Section 6 draws attention to the classification-oriented outstanding research challenges and hints towards the research gaps for future researchers. Finally, Section 7 concludes this paper.

## 2. Characteristics of dual high-penetrated power systems

This section summarizes the device and system-level features introduced by the dual high-penetrated power system.

### 2.1. Device-level features

Compared with the traditional power system involving classical TGs, the individual characteristics of CIGs (solar, wind, etc.) and other power electronic devices/equipment are mainly determined by the dominance of converter controls, multi-time scale dynamic response, low inertia, weak immunity, low short circuit ratio, obvious nonlinearities, and resource intermittency.

#### 2.1.1. Dominance of converter controls

The power electronic converter control plays a key role in defining the device-level characteristics. The grid connection of power electronic devices is driven by multiple control loops, such as the outer control loop, inner control loop, and pulsed-width modulation (PWM). Moreover, the grid synchronization of converter-based equipment is achieved through the phase-locked loop (PLL) and other similar mechanisms. Thus, the 'control algorithm' rather than physical characteristics of the device determines the response characteristics and control ability in the steady-state, dynamic state, and fault conditions [11].

#### 2.1.2. Multi-time scale dynamic response

The internal multi-level control of converter-based equipment contains multi-time scale coupling dynamics. For example, the double-fed induction generator (DFIG) wind turbine includes the controlled dynamic behaviors of ac voltage, dc voltage, and mechanical speed with the corresponding time constants of 10 ms, 100 ms, and 1s, respectively [12]. Thus, externally, it can respond to grid side disturbances in a wide frequency band, resulting in multi-time scale control interactions and new stability problems [11,13].

#### 2.1.3. Low inertia

Compared with the TGs, the inertia of the CIGs is low. Due to the power electronic converter interface, the input power of the prime mover is almost decoupled from the output electromagnetic power of the grid side. And thus, it no longer has the traditional inertia response characteristics based on rotational kinetic energy. The energy storage elements (inductance/capacitance) have a weak ability to absorb and discharge the energy response power deviation after being disturbed and resist the frequency deviations on the grid side. Quantitatively, for example, the inertia time constant of a 1000 MW thermal-powered TG is about 8–10s, while the inertia of grid-connected photovoltaic of equivalent capacity is almost 0 [14]. Although several control-based frequency support or virtual inertia technologies have been proposed [15], due to the lack of lasting energy support, their functions are limited and can affect the efficiency and flexibility of the equipment [15].

#### 2.1.4. Weak immunity to frequency and voltage deviations

Another dominant characteristic is weak immunity of the power electronic equipment with low overload capacity and insufficient tolerance to frequency and voltage deviations. In contrast to the TGs, the CIGs do not have similar primary frequency control and voltage regulation characteristics. Moreover, generally, the ability of the converter's, e.g., insulated gate bypass transistor (IGBT), ability to withstand voltage fluctuations is limited due to its rating. Usually, the rating of the IGBTs is just enough to withstand only small voltage fluctuations as the overall cost of the equipment is kept minimum by the manufacturer. For example, the upper limit of frequency and voltage tolerance of wind turbine generators is 50.2 Hz and 1.1pu respectively, which is much lower than that of 51.5 Hz and 1.3pu of conventional TGs. This makes the CIGs easy to get disconnected from the grid upon the large fluctuation of system frequency or voltage, which brings adverse impacts on system stability [15,16].

#### 2.1.5. Low short circuit current

The short circuit current of the power electronic equipment is small. Compared with the TGs, there is no sub-transient process in power electronic equipment during a short-circuit fault. The initial short-circuit current is low, usually within the range of 1.5pu [10], but the steady-state short-circuit current decreases rapidly due to high internal reactance and fast controllable reference current [17]. The amplitude and time-span of short-circuit current are smaller than those of traditional TGs, which brings challenges to control and protection equipment.

#### 2.1.6. Obvious nonlinearities

Compared with the traditional TGs, the nonlinear characteristics in CIGs are more obvious. The nonlinearities mainly come from discrete switching operation, control nonlinearities e.g., control limiters and saturation triggered due to overloading of CIGs, and sequential switching of control/protection strategies under different control modes (P/Q, V/f) or operating conditions such as low voltage ride-through (LVRT) and high voltage ride-through (HVRT) [18–20].

#### 2.1.7. Resource intermittency

The power generated by the TGs depends on controllable primary sources such as coal and water, which are stable and determined. In contrast, the input resource of CIGs relies on environmental and climatic factors such as wind speed and solar irradiance, which is continuously changing [21]. The wind and solar powers can be forecasted, but with significant uncertainties.

### 2.2. System-level features and challenges

The high-penetration of 'renewables and power electronics' introduces new system-level features and challenges, which are different from the traditional power system.

#### 2.2.1. Multi-energy and grid connections

Due to the increase in the proportion of renewable energy generation and new transmission/distribution technologies, there has been an increasing trend of connecting multiple electric power grids around the world, particularly in Europe. The multi-energy grid connections, multiple transmission/distribution networks e.g., ac and dc interconnections, and the coexistence of multiple power producers and market entities have led to significant changes in the system topology and stability characteristics.

The proportion of power electronic interfaced generation and load has increased sharply. Thus, the factors determining the dynamic behavior of the system have increased exponentially, in particular, the interactions between complex and diverse controls, which not only transform the classical stability characteristics but also introduce new stability problems.

#### 2.2.2. System operation and regulation

Since the size and capacity of CIGs are much lower than that of the traditional TGs, i.e. the number of CIG units with the same size is much higher, which leads to a significant increase in the number of unit combinations and operation modes. This brings great challenges to system operation and regulation.

#### 2.2.3. Security and reliability

The scale of onshore and offshore HVDC transmission has grown significantly. The capacity of a single project has reached 12 TW [22]. Meanwhile, the proportion of conventional TGs at the sending and receiving ends have reduced, which makes the frequency stability and transient voltage stability issues more prominent. Besides, the power loss caused by commutation failure in the HVDC transmission is huge, which significantly impacts the security and reliability of the system [23].

#### 2.2.4. Stability and control

With the increased share of CIGs and power electronic converter-based equipment, the overall system dimension increases, which makes the stability analysis and control more difficult. In addition to this, the individual device-level features such as the low inertia, weak stability, weak disturbance immunity, and generation intermittency of CIGs pose a new major challenge to the power system stability [11,12,15]. At the same time, the power electronic fast control ability also brings new opportunities and choices for stability analysis and control, which will be discussed in detail in Section 6 [24].

#### 2.2.5. Multi-time scale and wide-band electromagnetic dynamics

The multi-time scale and wide-band electromagnetic dynamics between the various converters and the traditional power grid components cause new stability problems [25–30]. Moreover, with the increase of system dimension and the exponential growth of combination mode, the difficulty of stability analysis and control increases sharply [31].

Here, it is worth mentioning that renewable power generation and power electronic control technologies are in the process of rapid development, for example, the transformation of grid-connected converters from 'grid following' to 'grid forming'. The grid codes or standards are constantly being modified and improved, which will continue to affect the future power system development trend.

## 3. Influence of dual high-penetrations on power system stability

### 3.1. Influence on classical stability problems

The proportion of renewable power generators and power electronic converters in today's power system has increased tremendously. The dynamic responses of renewables and power electronic equipment are different from the traditional power system equipment, which 'reshape' the overall dynamic behavior of the system and cause new stability problems [25–30]. On the other hand, the power injected into the grid changes the system operation mode, power flow distribution, and operating points of the traditional equipment, which as a result affects various aspects of classical stability, including power angle, voltage, and frequency stability [32,33].

#### 3.1.1. Influence on rotor angle stability

Low-frequency oscillation characteristics are mainly determined by the power grid structure e.g., inertia distribution, and operation mode e.g., amount of power flow. The large-scale grid connection of CIGs changes the power grid structure and power flow distribution at the same time, thus affecting the electromechanical oscillation characteristics [34–36]. The positive or negative influence and its extent depends on the penetration ratio, type, location, strength, operation, and control strategy/parameters (PTLSOC) [8], which changes the frequency, oscillation mode, and damping of the oscillation, and even introduce new oscillation modes [37,38]. Besides, the frequency of the low-frequency oscillation could even go beyond the known frequency range, which is 0.01–2.5Hz [38].

#### 3.1.2. Influence on transient angle stability

Transient angle stability is mainly determined by fault conditions (type, location, sequence), system structure, and power flow distribution before and after the fault. Since the increased proportion of renewable energy generation and power electronics changes the power grid structure and power flow distribution, it affects the overall transient angle stability of system. A large number of studies have shown that the level of influence depends on PTLSOC [39–43]. The special control modes, LVRT, and HVRT capabilities of the CIG also have certain impacts on the transient angle stability [44].

#### 3.1.3. Influence on voltage stability

The voltage-reactive power response characteristics of the CIGs and power electronic converter-based equipment in steady and transient processes determine their influence on voltage stability. In addition to the aforementioned PTLSOC factors, the influence of voltage-reactive control strategy on voltage stability is particularly obvious [45]. If the control is appropriately designed, the voltage stability can be improved

[46,47]. When the penetration ratio of CIGs increases, their connection with the grid becomes weaker, making the voltage stability control very challenging. Besides, due to the fast response characteristics of power electronic equipment, short-term voltage stability will include transient overvoltage/under-voltage and post fault recovery overvoltage/under-voltage [42].

#### 3.1.4. Influence on frequency stability

The performance of speed regulation/frequency modulation control and inertia are the key factors determining the frequency deviation and frequency nadir, i.e., the lowest value of the frequency after the disturbance. The CIGs lack the traditional 'inertia'. The fast primary-frequency support and smaller differential adjustment characteristics are subject to specific control strategies and economic factors, such as the maximum power point tracking (MPPT) control algorithms. Several studies have shown that, generally, the CIGs weaken the frequency stability of the system, resulting in higher frequency deviations in the steady-state, a larger rate of change of frequency (ROCOF), and lower frequency nadir [47,48]. One of the development trends is to improve frequency stability by upgrading the grid code specifications, for example, adding virtual inertia, frequency modulation control, and adopting grid-forming converter control [15]. The control flexibility of HVDC can also be exploited and additional control functions can be incorporated to improving frequency dynamics and reducing frequency fluctuation and offset [49].

#### 3.1.5. Summary

The influence of dual high-penetrations on the classical power system stability is summarized as follows:

1) There are significant differences in physical structure, mechanical characteristics, energy conversion mechanism, and control characteristics between the CIGs and traditional TGs, which are some of the major reasons for the change of stability characteristics.
2) The impact of dual high-penetrations largely depends on the PTSLOC of renewable power generators and power electronic converter equipment [50,51].
3) The development of grid codes plays a critical role in the evolution of system stability characteristics to formulate or update the grid code specifications according to the development of the system, and then guide the development of grid connection technology [43].
4) The proportion of renewable energy and power electronic equipment determines the extent of their influence on system stability. When the proportion of renewable energy and power electronic equipment is relatively low e.g., less than 1/3, it is necessary to consider its influence on the system stability determined by traditional equipment. When the proportion is equal e.g., 1/3–2/3, it is necessary to study and coordinate their dynamics to improve the overall stability. In the future, when the proportion becomes relatively high e.g., more than 2/3, there is a possibility of the influence of traditional equipment on the stability of the system and should be studied accordingly.

### 3.2. Emergence of new stability issues

The dual high-penetrations scenario also makes the power systems vulnerable to emerging oscillatory instabilities. The emerging stability problems in the dual high-penetrated power systems that are difficult to be classified into the 'classic' stability issues include 1) electromechanical-like low frequency oscillation, 2) electromagnetic wideband oscillation, and 3) new large disturbance stability problems.

#### 3.2.1. Electromechanical-like low-frequency oscillation

CIGs mostly adopt the PLL for grid synchronization. Without the traditional sense of 'rotor angle', the 'power angle stability' or 'electromechanical oscillation' issues cannot be studied directly. But this does not rule out that the CIG's output current or node voltage does not have synchronization or angle stability problems. The analysis in Ref. [52] shows that a new low-frequency oscillation mode may be introduced when CIGs are connected, and the oscillation frequency can be either low or high. In Ref. [53], it is pointed out that the PLL of DFIG and permanent magnet synchronous generator (PMSG) has the angle and differential variables similar to the traditional TGs, and has the second-order dynamics similar to the swing equation. This gave rise to the new 'angle stability' or 'electromechanical oscillation' problem and introduced the concept of 'PLL oscillation'. When a power electronic converter is connected to the ac power grid with a low short-circuit ratio, the oscillation frequency is higher than the traditional low frequency. However, if the damping of the oscillation mode is weak and falls into the frequency range of the traditional electromechanical oscillation, it participates in and affects the stability of the latter. Ref. [52] concludes that wind turbines and static synchronous compensators (STATCOMs) could also participate and affect the traditional electromechanical oscillation stability. The study showed that when the parameters of the STATCOM control did not match with the wind turbines', their interaction caused reactive power oscillation with a frequency of about 5.5 Hz. Note that the frequency is higher than the traditional frequency range of the electromechanical low-frequency oscillation. Refs. [13,25,54,55] reported that weak/zero damping low-frequency oscillation of 3~4 Hz occurred in weak ac grid in the research of electric reliability council of Texas (ERCOT) CREZ transmission project.

To sum up, the physical quantities reflecting these oscillations are the amplitude of the fundamental voltage or power, while the occurrence conditions are mostly related to two factors: 1) high level of power transmission, and 2) low short-circuit ratio. This is similar to the traditional low-frequency electromechanical oscillation, but its mechanism is quite different from the latter.

#### 3.2.2. Electromagnetic wideband oscillation

The characteristics introduced by the dual high-penetrations such as the dominance of converter control, wideband/multi-time scale dynamics, complex interactions of different equipment make the system stability no longer limited to the traditional fundamental frequency and electromechanical range. Instead, it triggers new types of electromagnetic oscillation with the frequency range extending from $10^{-1}$ to $10^{3}$ Hz, hereinafter called the wideband oscillation.

According to the frequency range, the new type of electromagnetic oscillation can be divided into sub-/super-synchronous oscillation ($S^2$SO) and {inter}-harmonic resonance/oscillation. Although these wideband oscillatory stability issues have been reported in the literature widely but are difficult to fit into the 'classical' stability category.

*3.2.2.1. Sub-/super-synchronous oscillation ($S^2SO$).* The classical subsynchronous resonance/oscillation (SSR/SSO) phenomenon is caused when "*the electrical network exchanges energy with the TG at one or more natural frequency of the combined system, below the synchronous frequency of the system*" [56]. The triggered oscillation is in the subsynchronous frequency range and is widely known as the SSO. In literature, the SSR and SSO terms have been used interchangeably. Note that the classical SSR/SSO problem is associated with the traditional TGs.

The SSR/SSO in CIGs and other power electronic converter-based devices also occurs [57]. The power electronic converter control actively participates in the interaction phenomenon and triggers oscillation in the subsynchronous frequency range. For this reason, the interaction is widely known as subsynchronous control interaction (SSCI). The oscillation triggered due to the SSCI is called the SSO if the frequency coupling effect or the super-synchronization is not obvious, and the sub-/super-synchronous oscillation ($S^2$SO) when the frequency coupling effect is obvious. When the oscillation occurs, the voltage or current waveforms contain non-characteristic components with a frequency ranging from several Hz to twice the fundamental frequency. The amplitude of the SSO/S$^2$SO continues to increase or even exceeds the amplitude of the fundamental frequency component under adverse conditions. Such diverging oscillation may cause frequent unit tripping and/or equipment damage, endangering the safe and stable operation of the power system [57–59].

The earliest SSO event occurred in a DFIG-based wind farm connected to a series compensated network in south-central Minnesota, USA, with an oscillation frequency of 9.44Hz [60]. Later on, the ERCOT system in the US (in 2009 and 2017) [61,62] and the Guyuan system in China (from 2012 to 2015) [26,63] reported several similar SSO events, with the frequency about 20–30 Hz and 3–12 Hz, respectively. On July 1, 2015, the Hami wind power base in Xinjiang, China reported an S$^2$SO event caused by the interaction between PMSGs and the weak ac grid [27]. The event caused subsynchronous oscillation with 27–33 Hz frequency as well as strong coupled supersynchronous oscillation with 67–73 Hz frequency. Somehow, the subsynchronous frequency matched with the natural torsional modes of the TGs operating hundreds of kilometers away. The torsional protection relays operated and tripped three of the TGs, and caused significant power shortage as well as increasing the risk of system-frequency drop. Not only do the wind turbines cause the S$^2$SO [64–67], but also the photovoltaic system has been reported to have subsynchronous frequency self-excited oscillation [68] or sub-harmonic oscillation [69]. The S$^2$SO caused by the interaction between STATCOMs and the power grid has been reported in Ref. [70]. In 2019, the UK's national grid experienced a significant loss of power generation in which the converter controls of the Hornessa offshore wind farms also participated and triggered SSO [71].

The classical SSO and the abovementioned emerging SSO/S$^2$SO are essentially different from each other. The SSR involves the torsional dynamics of conventional TGs. On the other hand, the mechanism of emerging SSO/S$^2$SO involves the dynamic interaction among the CIGs, converter-based equipment (FACTS/HVDC converters), and ac/dc power grid. The frequency, damping, or stability of oscillation depends on several parameters of converter control and power grid, as well as external conditions such as wind speed and solar irradiance.

*3.2.2.2. {inter}-harmonic resonance/oscillation.* The TGs are not sensitive to the high-frequency dynamics from the power grid due to their structure and parameter characteristics, such as large mechanical rotors, large time constant, and narrow passband. However, the CIGs and other power electronic equipment (such as the modular multilevel converter (MMC) based HVDC links) are sensitive to wideband dynamic response including medium and high frequency. Their interaction with the power grid may lead to the oscillation/resonance with frequency ranging from 100 Hz to more than 1000 Hz, resulting in harmonic overvoltage, overcurrent, and other serious power quality and system stability problems [69]. Several harmonic instability events have been reported around the world, including: 1) harmonic instability (high-frequency harmonic resonance) in the 100–1000 Hz range of Borwin1 offshore wind power project in the North Sea of Europe [72,73], 2) harmonic oscillation after photovoltaic power station failure in California [74], 3) high-frequency harmonic oscillation of ~1270 Hz in China's Luxi HVDC project [30,31], 4) 800 Hz harmonic resonance caused by offshore direct-drive wind power through ac grid connection [75], 5) 20/21st harmonic (1050 Hz) amplifications in Saihanba/Dongshan wind farms in Inner Mongolia and 21/22nd harmonic (1050/1350 Hz) amplification in a 50 MV A photovoltaic power plant in Qinghai [76], and 7) medium and high-frequency resonance (150–200 Hz) caused by measurement filter of photovoltaic grid-connected system under weak grid conditions [77], 8) harmonic resonance and instability caused by parallel connection of multiple photovoltaic converters with LCL filter into power grid [78,79].

Although the S$^2$SO and harmonic resonance instabilities have their distinct characteristics, frequency ranges, and dominant components, they are caused by the interactions between power electronic converters and the power grid, and the occurrence mechanism, analysis methods, and suppression techniques are similar. As field experience in Ref. [80] has shown, power electronic converters may produce wideband oscillation from sub-/super-synchronous to the medium and high-frequency range when it is connected to a weak ac grid [81,82]. In addition to the power electronic converter as an active device, passive grid components such as long transmission cables [83], passive filters [84], transformers [85], and series compensation capacitors [85,86] also play an important role. The emerging electromagnetic wideband oscillation seriously threatens the equipment safety, system stability, and power quality of the grid. These emerging oscillatory instability issues have become one of the major factors restricting the efficient consumption of renewable energy.

### 3.2.3. New large-disturbance instabilities

Under the background of a dual high-penetrated power system, there are numerous reports on the stability problems triggered by large disturbances such as faults. Although, most of the newly reported stability issues can still be classified into the category of classical stability issues. For some new problems, it needs to be further examined whether their mechanism is different from the classical problems or not.

*3.2.3.1. Large disturbance synchronous stability of CIGs.* In [87,88], it has been pointed out that the dynamic behavior of CIG's PLL is similar to that of TG's rotor. The large signal or transient stability of PLL can be studied using the equal-area criterion. Under high impedance ratio and LVRT or voltage sag followed by a large disturbance or fault, the interaction between PLL and ac current loop may lead to system instability. Ref. [89] points out that when a severe fault occurs, that is, when the voltage drops to a very low value, say below 0.2pu), a large reactive current e.g. 1.0pu is required for LVRT. The situation leads to a mismatch between output current and grid impedance characteristics, meaning that there is no steady-state operating point. As a result, the PLL is unable to track the system frequency, which leads to a loss of synchronization of the CIGs with the grid. Thus, the CIGs will not operate normally after the fault disappears and will be tripped by the protection system.

The above scenario is related to the synchronous operation of CIGs followed by a large disturbance. This kind of instability can be compared with the power angle stability of the TGs; however, the mechanism is very different. Ref. [90] mentions that the large-scale photovoltaic power plants in California, the United States were disconnected from the grid due to PLL and dc dynamics. The instability triggered by the PLL and dc dynamics was not expected in the traditional transient stability analysis studies.

*3.2.3.2. Electromagnetic transient aspects of post fault voltage.* Ref. [25] reported a short-term overvoltage instability caused by a weak ac grid in the research of the ERCOT CREZ renewable energy transmission project in the USA. Ref. [91] also pointed out that when CIGs are connected to a large-scale centralized connection, especially with the weak ac grid, problems such as high dynamic overvoltage, voltage instability, and severe voltage flicker may be caused. In the California power grid, steady-state overvoltage and post fault transient overvoltage problems were observed during the operation of photovoltaic power stations, which led to transmission/distribution line overloading, overvoltage tripping, and even system-level transient instabilities [74]. It should be pointed out that most of the existing stability studies in the context of dual high-penetrated power systems focus on small disturbance and oscillation in the PLL-based grid-connected CIGs. The research on non-PLL grid-connected CIGs, large disturbance, and transient stability is relatively less. However, weak immunity, small short-circuit current, and discreteness of CIGs will show different response characteristics with TGs under large disturbances, which will have a significant impact on the stability of large disturbances.

Overvoltage problems caused by power electronic equipment and their control have also been widely concerned, such as 1) overvoltage caused by MMC control [92], 2) overvoltage oscillation caused by the propagation of high-frequency PWM pulses through cable [93,94], 3) crosstalk overvoltage between dc and ac sides of PV inverter caused by the photovoltaic array to ground capacitance [95], 4) resonance caused by converter control or frequency conversion speed regulation in traction power supply system [96], and 5) dc side overvoltage of converter [97]. These overvoltage problems are quite different from the traditional voltage stability problems. Most of them are related to electromagnetic processes and affect the safe operation of the equipment. Under adverse conditions, the power electronic converted based equipment may get disconnected from the grid or even damaged, endangering the stability of the power grid.

## 4. Discussion on existing power system stability classifications

### 4.1. Overview and historical background

The existence of standard definitions and classifications is the basis of power system stability research, which has always been the focus of the relevant organizations and authorities.

In 1937, the American institute of electrical engineers (AIEE) issued the first report on power system stability [98], focusing on the analysis of the synchronization problem of parallel operating of synchronous generators. The problem was perceived as a stability problem and it was one of the important concerns of the engineering community at that time. The power system stability was divided into two categories: 1) steady-state stability, 2) transient stability.

CIGRE published a series of technical reports on the definition and classification of power system stability in the 1950s, 1960s, and 1970s [99–101]. Subsequently, the IEEE Power System Engineering Committee established a special working group for the 'terminology and definition', which recommended terms and definitions of power system stability in 1982 [102]. The task force proposed a new classification method and divided it into two main categories:

- **based on the disturbance size:** power system stability was classified into 1) small disturbance stability (replacing the previous steady-state stability) and 2) large disturbance stability.
- **based on the time span:** power system stability was classified into 1) short-term stability and 2) long-term stability.

The power system instability was also classified into 'monotonic instability' and 'oscillation instability' depending on whether the instability was caused by insufficient synchronous or damping torque. The voltage instability caused by insufficient reactive power support was also mentioned, and the classification methods of voltage instability and power angle instability were also proposed.

In 2004, the IEEE/CIGRE Joint Working Group on stability terms and definitions published 'definition and classification of power system stability' [9,103]. The report has been widely used and has become the basis for discussing the expansion and classification of stability concepts. The power system stability was classified into:

- **three main branches:** based on the system variables that characterized the physical nature of the instability mode and in which instability can be observed: 1) rotor angle stability, 2) voltage stability and 3) frequency stability, as shown in Fig. 1.
- **two subbranches for each main branch:** according to the disturbance size (small/large) and time-span (short/long) needed to be considered in evaluating the stability.

The aforementioned reports on the terms, definitions, and classification of stability mainly focused on traditional large-scale power systems. Since the 21st century, the rapid development of power electronic and microgrid technologies has brought new stability problems. The IEEE established a working group that published its final report in 2020 on the definitions and classification of power system stability [104,105]. The authors of this paper show that different perspectives can be adopted for the definition and classification of stability for different power systems.

### 4.2. Discussion on the IEEE/CIGRE classification of 2004

As a background, this subsection critically assesses the applicability and adaptability of the definition and classification of stability provided by IEEE/CIGRE in 2004.

- The angle, voltage, and frequency terms in the classical stability refer to the angle, voltage, and frequency of the fundamental component. This implies that the classical stability definitions and classification are limited to stability issues associated with the fundamental components and do not have a clear definition for the power system stability issues associated with the signal components other than the fundamental frequency.
- The angle in the classical definition of stability is the rotor angle of the traditional TGs. Thus, the classical rotor angle stability is no longer applicable for electric power generators without physical rotor/angle such as the PV power generator.
- The voltage in the classical definition of power system stability, although it is not explicitly stated, is perceived as the RMS value of fundamental component from the signal definition, separating it from angle and frequency, and stability analysis practice.
- Some typical stability problems have been 'embarrassing' the 2004 classification for a long time, which are difficult to be integrated. For instance, the instability caused by the SSR/SSO, which appeared in the early 1970s and caused widespread concern [106,107], has been neglected in many discussions and literature on power system stability and classification. In some references, it has been simply and roughly classified into the category of small disturbance angle stability [108]. Ref. [108] states to refer to the conclusion in Ref. [109], which is not quite accurate because of the following reasons.
- The power angle stability has been regarded as an electromechanical stability problem for a long time, while the SSR/SSO is an electromagnetic stability problem – one adopts an electromechanical model while the other adopts an electromagnetic model.
- The power angle stability considers the angle of the fundamental component only (the relative motion between the rotor as a whole and the rotor of other units as a whole). While, the SSR/SSO is associated with sub/supersynchronous frequency components, which exist independently relative to the fundamental component. The SSR considers the torsional vibration between the shaft sections of the unit and its induced sub/supersynchronous voltage and current variations.
- The SSR includes transient torque amplification and other transient processes, which also opposes its classification in the small disturbance category.

The classical stability classification of 2004 is the product of the conventional power system era, which was mostly dominated by the traditional TGs. With the dual high-penetration trend of modern power systems, as mentioned above, many emerging stability problems cannot be incorporated into the classical stability classification. Just like the previous historical development of stability definition and classification, it is now a critical time to reexamining and revisiting the definition and classification published in 2004.

### 4.3. Discussion on the IEEE classification of 2020

To address the challenges and account for the emerging instabilities, a new IEEE Working Group was formed. It released a technical report titled “stability definitions and characterization of dynamic behavior in systems with high-penetration of power electronic interfaced technologies” in April 2020 [104]. The working group proposed a new stability classification, as shown on the right side of Fig. 1. For the most part, the proposed stability classification of 2020 retained the existing stability definitions of 2004 [9] and merely added two new branches, that is, resonance stability and converter driven stability.

- **Resonance stability:** It includes electrical resonance and torsional vibration. *Electrical resonance* refers to the electromagnetic oscillation caused by the dynamic interaction between the CIG and the power grid in the purely electrical sense such as the induction generator effect (IGE), which was classified as SSR in the early stage [56], and as SSCI recently [58]. The *torsional vibration* mainly refers to the oscillatory instability caused by the interaction between the mechanical system of the rotating unit and the ac electrical network with the series compensation and other converter-based equipment, such as static var compensator (SVC)/STATCOM. Thus, it includes the classical SSR and device-dependent SSO but excludes the IGE.
- **Converter-driven stability:** The multi-time scale control characteristics of the CIGs lead to the coupling interaction of electromechanical dynamics and electromagnetic transients between the CIG and the network, resulting in the oscillation in a wide frequency range. Based on the frequency, the *converter-driven stability* is further divided into *slow interaction* and *fast interaction.* The former has a lower frequency, typically less than 10 Hz; the latter has a high frequency, typically from tens to hundreds of Hz, even up to kHz.

The newly added branches (terms and classification) in the extended stability classification of 2020 [104] reflect the influence of dual high-penetrations in the modern power system, which intended to address the limitations of the classical classification of 2004. However, the 2020-classification is extended by just ‘patching’ some new terms to cover the emerging stability issues. According to the understanding of the authors of this paper, the extended classification of 2020 lacked rigor and left some of the issues unaddressed, which are summarized in the following points.

- The ‘logic’ and ‘scale’ of the classical classification method (left part of Fig. 1) are confused by the extended classification, as shown in the right part of Fig. 1. The original stability classification of 2004 was organized into three layers: 1) system variables, 2) disturbance size, and 3) time-span, which can coexist but can easily be distinguished. In this sense, the extended classification of 2020 is inconsistent with the classification methodology of the classification of 2004, which could be observed in the left and right parts of Fig. 1. For example, the extended classification of 2020 places the
- system equipment (such as CIGs, rotor shafts, etc.) and stability phenomenon (resonance stability) in the *system variable layer* (i.e. rotor angle, voltage, and frequency) of the classical classification of 2004.
- dynamic speed (slow or fast interaction) and subclasses of the resonance stability (such as electrical and torsional resonance) in the *disturbance size layer* (i.e. small and large disturbances).

The inconsistency of the classification methodology or layers between the classical and extended classifications creates confusion and makes it difficult to comprehend and understand the emerging stability problems.

- The ‘resonance stability’ and ‘converter-driven stability’ issues are hard to distinguish from each other. This is because the resonance stability could also be driven by converters-based devices.
- The extended classification does not completely solve the ‘inadaptability problem’ of the classical classification under the background of dual high-penetrated power systems. For example, in the extended classification of 2020, the ‘rotor angle stability’ does not cover the stability issues involving CIGs. But in fact, the CIGs do participate and affect the synchronous stability of the system. Therefore, it is unreasonable to completely exclude CIGs from the independent portal of the rotor angle stability.
- The extended classification is not fully inclusive. For example, it does not cover some of the actual stability phenomena that have been reported in various real-world power systems around the world. For instance, the oscillation phenomena of 2.5 Hz + 97.5 Hz in China, the short-term electromagnetic transient voltage problem due to faults in dual high-penetrated power system [25,91–97] are very difficult to be classified into a subclass of Fig. 1.

## 5. Proposal of new stability classification and framework

### 5.1. Background

The electrical power system is a large-scale complex dynamic system. The term ‘oscillation’ is used to describe its internal variables (such as voltage, current, and power), components (such as generators, transformers, etc.), subsystems, and the overall periodic variation characteristics and/or dynamic processes. Traditionally, the system frequency of alternating current-based power systems around the world is 50 Hz and/or 60 Hz referred to as ‘power frequency’ or ‘fundamental frequency’, including a small amount of 0 Hz direct current components. Ideally, the power system generates, transmits, and supplies 50 Hz or 60

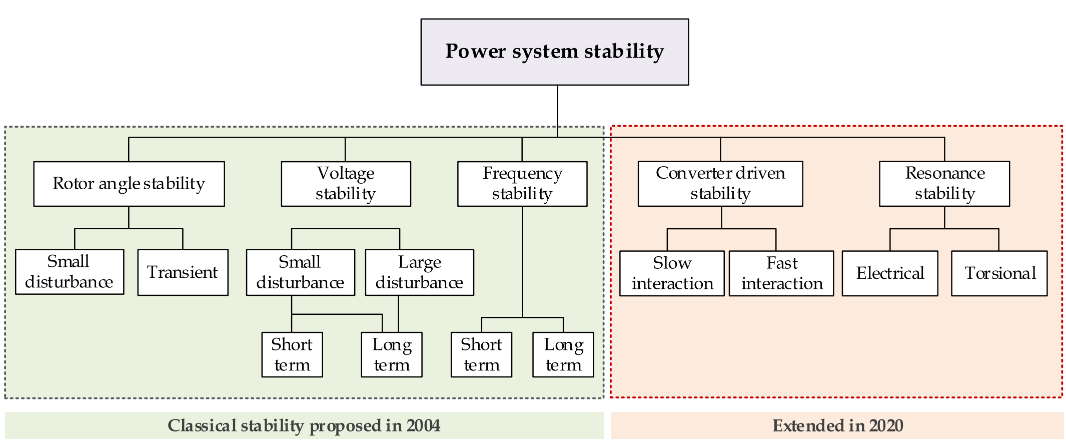


**Fig. 1.** IEEE classifications of power system stability (Year: 2004 and 2020).

Hz (plus a small amount of 0 Hz dc) voltage and current. In this context, the term "power system oscillation" usually refers to the occurrence of dynamic energy exchange (mechanical, electromagnetic, etc.) process of its variables, components, and even the whole subsystems under specific conditions. This dynamic energy exchange is considered 'parasitic' in the normal operation of the system, as it could potentially cause system stability or power quality problems. Therefore, the 'power system oscillation' is unfavorable to the smooth operation of the power system. It needs to be prevented, avoided, and suppressed in power system planning, operation, and control.

The 'oscillation' can be divided into 'fundamental component' and 'non-fundamental component'. For the safe and stable operation of the power system, the former is '**necessary to be maintained**' at a specific level (50/60 Hz), and the latter is '**necessary to be avoided**'.

The specific term 'oscillatory stability' refers to the power system characteristic of whether the dynamics of the oscillation can converge under a small or large disturbance. If the oscillation amplitude continues to increase or persists, it is unstable; on the contrary, if its amplitude is within the allowable range and gradually converges, it is stable. The oscillatory stability can usually be judged based on the damping of the oscillation mode. It is one of the major technical challenges in modern power engineering research. With the fast development and growth of dual high-penetrated power systems, many new 'oscillatory stability' issues are being reported.

### 5.2. Proposed classification

This paper proposes a new power system stability classification framework, which has several advantages over the existing power system stability classifications of 2004 and 2020. The proposed classification is shown in Fig. 2. The proposed framework aims at helping researchers and engineers better understand, define, and classify the emerging power system stability issues in the context of the dual high-penetration scenario. Compared with the IEEE's classical and extended classifications of 2004 and 2020 as shown in Fig. 1, the authors propose three major changes in the classification, as illustrated in Fig. 2.

#### 5.2.1. Addition of top-level new layer for fundamental and non-fundamental component stability

A new layer is added, which classifies the power system stability based on the concerned frequency of the component, that is the fundamental component stability, and the non-fundamental component stability.

***Fundamental component stability*** is mostly dominated by electromechanical dynamics and refers to the ability of the system to maintain stable operation at the system's power frequency i.e., 50/60 Hz for the ac and 0 Hz for the dc side, and provide high-quality electric power to customers. Further terms, definitions, and subclassification such as angle, voltage, and frequency stability are consistent with the classical definition and classification proposed in 2004.

***Non-fundamental component stability*** is mostly dominated by electromagnetic dynamics and refers to the ability of the system to avoid or resist the unwanted inherent non-fundamental wave electromagnetic oscillation components. In addition to the classical SSR caused by torsional vibration or IGE, there may be other inherent electromagnetic modes of the system, which may converge rapidly and do not lead to stability problems. However, the converter-based equipment due to the customized and flexible control strategy may produce the 'negative resistance' effect in a wide frequency range, resulting in complex and time-varying non-fundamental frequency oscillation problems.

#### 5.2.2. Sub-categories for non-fundamental component stability

***Non-fundamental component stability*** is further divided into three subcategories according to the frequency range of the resonance/oscillation.

***Sub-/super-synchronous oscillation*** or $S^2SO$ includes the electromagnetic oscillations frequency ranging between 0 Hz and $2f_0$Hz, excluding 0 and $2f_0$, where $f_0$ is the fundamental frequency. The $S^2SO$ can be defined as the electromagnetic oscillation, with frequency within the sub-/super-synchronous frequency range, triggered due to either the classical SSR phenomenon associated with classical TGs or the SSCI phenomenon involving converter controls of a power electronic equipment, such as WTGs. In literature, the SSR, SSO, and SSCI terms have been used interchangeably to represent the electrical resonance at subsynchronous frequency.

***Medium/high-frequency resonance/oscillation*** includes the oscillation within $2f_0$Hz and 1 kHz, excluding the $2f_0$ for the medium-frequency oscillation, and more than 1 kHz for the high-frequency resonance. The *medium-* and *high-frequency oscillation* is sometimes referred to as *harmonic resonance/oscillation.*

The frequency band range of the above terms, especially for the medium- and high-frequency oscillation, is ambiguous and can be adjusted based on further research and real-world resonance/oscillation events.

Further in terms of disturbance size, the non-fundamental component stability is subdivided into a *large-disturbance* transient process excited by short circuit faults and *small-disturbance* near the operating point.

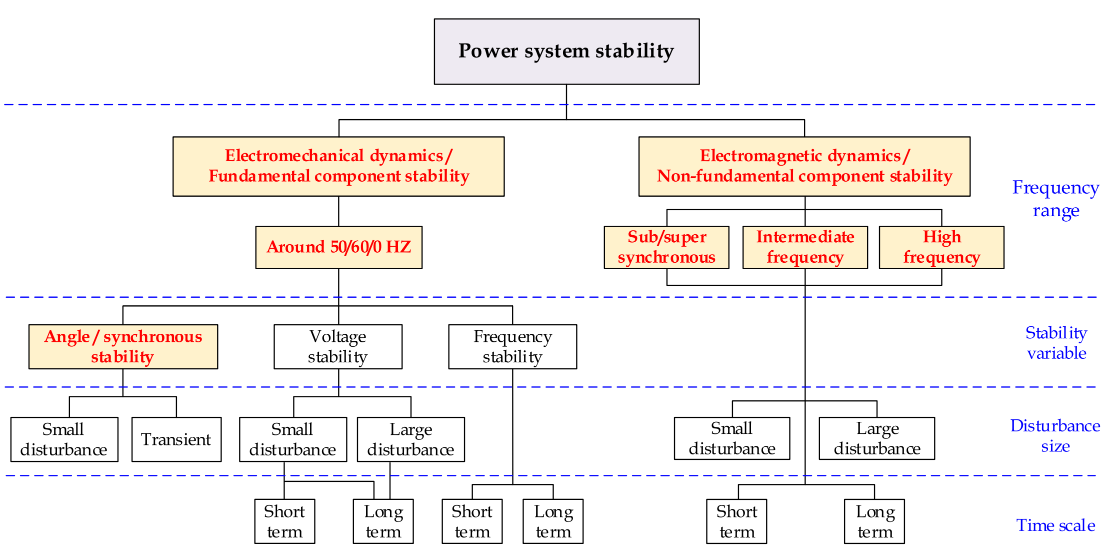


**Fig. 2.** Proposed power system stability classification.

*5.2.3. Replacing 'rotor angle stability' with 'angle/synchronous stability'*

The existing *rotor angle stability* in the classical classification is modified to ***angle/synchronous stability***, which now accommodates the classical rotor angle stability of synchronous generators as well as the angle of the equivalent internal potential voltage of the CIGs [12,110] or the angle of the point of common coupling (PCC) voltage. Thus, the *angle/synchronous stability* refers to the ability of both TGs and CIGs to maintain the synchronous operation with other equipment or the whole system.

### 5.3. Distinct features of the recommended classification

The recommended classification has the following features:

- It maintains the inherent logic of the classical classification method and has better continuity and coverage. It eliminates the layers' inconsistency between the classical and extended stability classifications of 2004 and 2020.
- A stable power system should be able to 1) maintain the system's fundamental frequency and voltage under various real-world operating conditions and disturbances, and 2) eliminate or suppress the non-fundamental components, such as the wideband oscillation. The fundamental component is the carrier of energy transmission, and its stability can be subdivided into angle, voltage, and frequency. However, the non-fundamental components should be suppressed and thus there is no need to divide it based on the stability variables, as reflected by the empty-space corresponding to the stability variable. The recommended classification better reflects these basic requirements.
- The frequency ranges of the stability problems have a direct influence on the order of the model and the complexity of the stability analysis. It is reasonable and beneficial to model, analyze, and control the stability by dividing the system dynamics and stability according to the concerned frequency band. For instance, the classical fundamental phasor method can be used for investigating the stability issues related to fundamental frequency; the electromagnetic transient or dynamic phasor method can be used for studying the stability issues associated with the non-fundamental frequencies. The proposed classification provides a better view and clearly distinguishes the instabilities associated with the fundamental and non-fundamental component stability.
- The recommended classification has better scalability and future adaptability. The essential characteristics of power system dynamics are highlighted, while the leading equipment is kept behind. For example, the recommended classification does not specifically highlight the leading equipment in classification, that is, the synchronous generator or converter equipment is not highlighted.

Here, it is pertinent to mention that the newly added non-fundamental frequency stability problems are mainly manifested in the non-fundamental frequency resonance/oscillation caused by the interaction of electric power generators (TGs and CIGs), electrical network, magnetism, and converter controls in the system. The difference and connection between the new non-fundamental frequency stability problems and the traditional inter/sub-harmonic power quality problems are that: the former is used to describe the inherent dynamic characteristics of the system, which is closely related to differential equations; the latter is a non-sinusoidal periodic component of voltage, current, and other variables. When the non-fundamental frequency oscillation mode of the system is excited, it may cause harmonic components as well as corresponding power quality problems in the voltage and current. On the contrary, when the harmonic components appear in the voltage and current, it does not necessarily mean that there is an inherent oscillation mode, it may also be caused by nonlinearities. For example, the nonlinear characteristics of some power sources or loads can cause voltage or current distortion harmonic generation.

### 5.4. On the classification of aperiodic instability

In the classical stability classification of 2004, aperiodic instability is classified in the electromechanical dynamic category. As described in Ref. [109], the aperiodic instability caused by insufficient synchronous torque under small or large disturbances belongs to the power angle stability category. In other words, the abovementioned division of *fundamental frequency stability* and *non-fundamental frequency stability* does not only refer to the frequency of *'oscillation'* but focuses on whether the dynamics are electromechanical or electromagnetic. Therefore, the aperiodic instability or lack of stable operating point after a disturbance can still be classified as the fundamental component stability.

### 5.5. On the boundary and overlapping of different stability subclasses

The stability of a power system is usually seen as a whole. However, under a specific scenario, a certain variable or characteristic is highlighted, which mainly reflects the stability of only a certain aspect. The stability classification is only to observe or express the system stability as a whole from different aspects. The stability of different aspects or categories is coupled, overlapped, and related. The classification does not decouple the system stability into uncorrelated parts. As mentioned in Ref. [109], classification deals with different aspects of stability problems. It is desirable to analyze the overlap of power system stability phenomena to some extent.

The overlap between subclasses is divided based on the 'variables'. For example, the power angle stability and voltage stability in the classical stability are essentially coupled. The *pulling-out* of power angle between units, at the same time, may also cause a decrease in voltage amplitude. It can be said that under some conditions, the dynamics between power angles of units dominate the stability, while the voltage dynamic is secondary, but the system is classified in the power angle stability category. In other cases, for example, if the reactive power imbalance is dominant after a fault, the voltage dynamics may be in the dominant position and the power angle dynamics are secondary, then the system's stability is classified as voltage stability.

The stability is divided into electromechanical or fundamental component stability and electromagnetic or non-fundamental component stability. The overlap between subclasses is categorized based on the frequency bands, which are fuzzy and hard to distinguish. For example, in the process of classical power angle or voltage or frequency instability, some non-fundamental components such as the $S^2SO$ can also appear in the voltage/current dynamics. Also, there are some fuzzy zones in the frequency range of low-frequency oscillation and $S^2SO$. The traditional low-frequency oscillation is the power oscillation with an oscillation frequency less than 2.5 Hz formed by local or interarea synchronous generator units or plant connected to the grid. As a result, the fundamental frequency component of the voltage and current will be modulated to produce $50\pm<2.5$ Hz components, overlapping with the sub-/super-synchronous frequency band. On the other hand, the frequency range of electromagnetic oscillation caused by the control interaction of power electronic equipment is quite wide. The inherent $S^2SO$ mode which is close to the fundamental frequency may be triggered under certain conditions, which will modulate the fundamental frequency signal and lead to a phenomenon similar to low-frequency oscillation. To solve the stability problem of this kind of frequency band overlap, one should not only identify the stability mechanism and classify it correctly but also fully realize the complexity and coexistence of the electromechanical and electromagnetic dynamic interactions. The aspects must be taken into account in the modeling, analysis, and control of both electromechanical and electromagnetic instabilities.

The overlapping of large/small disturbances and short/long-term stability subclasses has already been discussed in Ref. [109] and is repeated here.

### 5.6. On the system stability and equipment stability

When discussing the concept of power system stability, we generally focus on the stability of the whole system, but less on the stability of individual equipment. It should be noted that system stability and equipment stability are closely related, sometimes even difficult to distinguish. For example, the equipment stability of a single unit in a single machine infinite bus system, which is a common scenario of stability analysis, is the stability of the system instead of the stability of the single machine. But in the actual complex system, the instability of single equipment may also lead to a cascaded failure of multiple equipment, eventually leading to system instability. However, if individual equipment is unstable due to its local issues such as unreasonable parameter setting, control imbalance, it will not endanger the stability of the whole system. Instead, the equipment instability will be regarded as a *disturbance* to the overall system and can be distinguished from the system stability.

In practice, there is no universal and clear rule to distinguish *system stability* from *equipment stability*, which depends on the specific situation and operation requirements of the power system.

The stability condition defined in paper [102] is '*reaching an acceptable steady-state operating point after disturbance*', which also points out that the acceptability depends on the specific situation and operation standards. The stability condition of ref. [9] is '*most of the system variables are within the allowable range, so almost the whole system can remain intact*', which also points out that attention should be paid to the stability of specific generators and loads. In the latest definition of stability [10,104], it is explicitly proposed that *local instability* and *stability of a control loop* should not be considered. The expressions in the stability definitions use expressions such as 'acceptable', 'majority' and 'almost' and other 'ambiguous' words, which provide convenience for different power grids and their technical management departments to formulate operational rules according to their actual situation.

The device and system-level features of the power system with dual high-penetrations are the main reason behind the emerging stability issues, highlighting the need to revisiting the power system stability classification. For example, the dominance of power electronic converter controls and their multi-time scale dynamics cause the emerging wideband oscillation. The classification of 2020 in Fig. 1 tried to incorporate the emerging instabilities, caused by the device and system-level features of the power system, by merely patching with two new branches i.e., resonance stability and converter-driven stability. Based on the device- and system-level features of the modern power system with dual high-penetrations, it can be said the modern power system should be said as 'stable' if it can maintain stable operation at the fundamental frequency and ability to resist components other than fundamental frequency components introduced mainly due to device- and system-level features. The proposed classification of Fig. 2 gives a better view to incorporate this basic requirement.

## 6. Research prospects

### 6.1. Research prospects of fundamental component stability

#### 6.1.1. Research prospects of angle/synchronous stability

The angle/synchronous stability of the dual high-penetrated power system is associated with the synchronization of the CIGs with the grid. The angle/synchronous stability is mainly determined by the inertia distribution, amount of power flow, grid strength, etc. The future research directions are as follows.

- At present, the grid following control strategy is dominant for the grid-connection of CIGs. However, the converter control mode of the power grid is gradually changing from the grid following to grid forming. Different from the previous grid following converter, grid forming converter can provide reference values for frequency and voltage and directly adjust these values [111]. To effectively solve the low inertia problem of a dual high-penetrated system, a grid forming converter also needs to show load sharing, droop, and black start characteristics like the TGs. Unlike the TGs, this control mode does not initiate any physical synchronization and stabilization mechanisms, nor does it provide any physical inertial response. Therefore, the above-mentioned key functions of the synchronous machine must be realized by controlling the converter as well as independent energy storage. Although these functions may not be necessary for the future converter-based grid, the transition period is expected to be long, during which the TG and grid forming must be able to interact and ensure system stability.
- In the future power grid, the in-depth study of how to quantify the grid strength and choose the appropriate control structure accordingly, including the proportion of control structure and the impact point in the power grid. For example, reference [112] explores the stability of networking equipment in improving the access system of large-scale PLL-based power electronic equipment from the perspective of grid strength. The grid structure also plays a key role in the stability characteristics of the power system [113]. Therefore, how to reasonably plan the grid structure to reduce the risk of system instability is also the content to be studied.
- At present, the synchronous control structure of power electronic equipment is often obtained from engineering experience and physical intuition, which is also an important reason for the instability of some typical structures in specific scenarios. What kind of synchronization control structure is needed in the power system with a high proportion of power electronic equipment, and how to design the optimal synchronization control structure is still a problem to be solved.
- The amount of power flow also impacts the angle/synchronous stability. In future, the modern artificial intelligence and big data have to potential to find applications for accurate predictions of load-demand of flexible resources, and their intelligent scheduling and dispatching, which can eventually lead to a secure and reliable power system [114,115]. The traditional model-driven power system operation and control theories face challenges such as strong uncertainty, complex security mechanism, and insufficient flexibility. The traditional model-driven power system regulation theory cannot adapt to it. Thus, it is urgent to combine data-driven and model-driven approaches to create a new theory of intelligent regulation and control, to lead the operation from automation to intelligent type development [116]. The advanced artificial intelligence algorithms can handle complex security constraints such as managing the power flow such that the risk of oscillatory stability, including the angle/synchronous stability, is reduced [117]. An efficient solution method could propose optimization scheduling with complex high dimensional constraints based on bio-inspired learning algorithms [118,119].

#### 6.1.2. Research prospects of voltage stability

The voltage stability of the power system under the dual high-penetration scenario is faced with issues, such as cascading faults, dynamic interactions, strong nonlinearities, and intermittent resources. The research lacks the corresponding theoretical basis to investigate and mitigate the voltage stability issues. The research prospects in the following aspects should be carried out:

- Revealing the cascading failure mechanism of voltage stability and the resulting 'security risk' and propose an efficient identification method of key faults.
- Studying the dc control and protection system under different disturbances based on the interaction between complex fast control and large-scale grid-connected equipment. Moreover, a quantitative calculation method of the voltage safety region of the sending end should be proposed.

- Investigating the impact of strong nonlinearities such as control saturation and control limit of the dc system and multi-time scale dynamic reactive power control characteristics of the ac system.
- Aiming at the modern power systems with dual high-penetrations, intuitive ideas for the prevention or mitigation methods should be put forward, for example, the coordinated prevention and control to ensure voltage stability.

#### 6.1.3. Research prospects of frequency stability

The frequency stability of the modern power system under the dual high-penetration scenario is faced with the problems of low inertia, multi-time, and low tolerance. To solve the major problems of frequency stability, there is a need of carrying out comprehensive researches in the following directions.

- The differences in frequency response characteristics of various types of CIGs on different time scales should be studied. Moreover, the mechanism and characteristics of the time-space evolution of the low inertia system should be revealed.
- The fast calculation and real-time monitoring/warning of the key frequency stability indices should be proposed. The threshold of frequency dip and recovery time after disturbance should be evaluated in real-time under complex system conditions.
- The adaptive coordination control theory of frequency dynamics can improve the full-time frequency stability from 'inertial support' to 'three-stage frequency modulation'.

### 6.2. Research prospects of non-fundamental component stability

The *non-fundamental component stability* includes the *wideband resonance/oscillation instability*, which comprises SSR/SSCI and harmonic resonance phenomena in dual high-penetration power systems and is generally caused by the dynamic interaction of power electronic converters and their controls with the ac/dc grid.

The monitoring and control of wideband oscillation are faced with the problems of wide-spectrum frequency, time-variation, and nonlinearity [120]. The traditional techniques of prior modeling, offline analysis, and customized control of oscillation stability are not feasible. The innovative idea is to change the traditional oscillation stability analysis and suppression principle from '*fixed frequency, offline, and customized control*' to '*wideband frequency, real-time analysis, and intelligent/adaptive control*', as described below [121].

Traditionally, the synchrophasor technology is designed to measure fundamental phasors with a small reporting rate. However, modern synchrowave/synchrophasor measurement units (SMUs) are expected to record real-time voltage/current waveforms with time-stamps and a high reporting rate. Wilsun et al. [122] published a visionary paper, which highlights the applications of "synchronized waveforms" for monitoring, protection, and control of apparatus as well as system. Meegahapola et al. [123] also highlighted the potential of synchrophasor technology for monitoring, analysis, and control of emerging oscillatory stability. Here, the authors identify three research directions related to wide-spectrum real-time identification, monitoring, follow-up analysis, and control of emerging oscillatory stability.

- **Wide-spectrum identification**: Embedding broad-spectrum sensors in the power grid to realize fast and accurate identification and monitoring of wideband oscillation modes and frequency-varying model. For example [116], uses a multiple-support vector machine model, which extracts the features of the disturbance and selects the proper classifier among the multiple-support vector machine models. The proposed scheme for SSO identification was tested through field measured PMU data. Ref. [124,125] used an adaptive extended Kalman filtering approach for SSO monitoring in real-time, which can also be used for online detection of SSO events. Given that the synchronized waveform is measured, and information of wideband oscillation is preserved and identified, various algorithms can be developed [126,127].
- **Follow-up real-time monitoring and analysis:** Real-time tracking and follow-up stability assessment of the wideband oscillation [123]. Similar to the work of [116], new data analysis tools could be exploited for the online characterization of emerging oscillation [128]. Besides, the measured oscillatory waveform information can be used to compute the impedance model of power electronic equipment in real-time. The real-time analysis and control functions can be added in the control center to make an appropriate decision as a countermeasure to the instability. For example, the power flow associated with the multi-frequency oscillation can be determined, which in turn can be used to determine the source of the oscillation [129,130]. Ref. [131] proposed a distributed-SMU-based system--wide protection algorithm to mitigate the SSCI in a realistic large-scale wind power system. The impedance models of the wind farms were measured according to the sub-/super-synchronous frequencies and sent to the control center for real-time analysis. Based on the preset criteria and the computed sensitivity index, a certain number of wind farms were tripped until the oscillation was stabilized. The synchronized waveform measurements could be exploited for device-/system-level applications related to monitoring and analysis of emerging wideband oscillation.
- **Adaptive control of wideband oscillation:** In the actual system, the frequency of the wideband oscillation is determined by the system's operating conditions as well as the control configuration and parameters of the interactive converter-based equipment. The overall configuration and parameter design of the control system faces huge adaptability and robustness problems. It is necessary to consider the time variations in the oscillation frequency due to changing operating conditions. With the help of synchronized waveform technology [122,132], it is possible to detect and estimate multi-frequency oscillations. The adaptive control of wideband oscillation is based on the measured wideband frequency, the individual dynamic and overall interaction of the equipment should be adjusted adaptively. For example, in a recent study [133], the authors of this paper used a frequency adaptive damping control of SSO in DFIG-based farms connected to a series-compensated network. The online estimation of oscillation frequency and the corresponding countermeasure is necessary for the stable operation of modern power systems. In a practical sense, there is a lack of highly efficient practical device/equipment to suppress the non-fundamental/wideband oscillation, which is, as stated earlier, is one of the requirements of the modern power system to be said as "stable".

### 6.3. Modeling and stability analysis research prospects

The classical theory of power system stability is based on a synchronous generator and fundamental phasor model. Under the dual high-penetration scenario, the mechanism and characteristics of the power system stability change significantly. The modeling and stability analysis of power electronic equipment is more complex than traditional power systems because of its multiple control loops, strong coupling, complex grid-connected conditions, and strong nonlinearity caused by control switching/saturation. To study the power system dynamic behavior under the dual high-penetration scenario, the device or system-level modeling depends on whether the dynamics of concerned stability issues are associated with the utility/fundamental frequency or non-utility/wideband frequency. The applicability of existing equipment level modeling to study the emerging instabilities needs comprehensive research. At present, compared with the traditional power system dominated by TGs, there is no mature modeling method in describing the interaction between CIGs and other system equipment, and thus it is difficult to fully understand the mechanism and characteristics of the emerging power system stability issues. Since the complexity of modern power systems is increasing, new types of oscillatory problems are being reported. For some of the reported events, the root causes and/or the mechanisms behind those incidents are still unknown. For example, the mechanism and characteristics of the following problems are still not fully known with generic models.

- The high-penetration of renewables causes problems such as frequency and voltage stability, which are interlinked. The effect of local faults propagates and spreads to the whole grid, which poses a great threat to the security and stability of the power system.
- The operation of converter-based equipment under fault conditions often involves control nonlinearities such as limiters, switching functions, and delays. The incorrect control settings could lead to over-voltage/current problems, deteriorating the control performance under fault conditions. Since the converter-based equipment relies on control configuration and settings, understanding the instability mechanism has significant importance for the satisfactory performance of the converter-based equipment.
- The high proportion of power electronic converters leads to the prominent wideband oscillation problem dominated by converter control with oscillation frequency ranging from several hertz to several kilohertz. Several such wideband oscillation events have occurred in the power systems around the world, including the US, UK, and China.

#### 6.3.1. Modeling in time-domain

One of the barrios to detailed investigations of emerging unstable is the lack of availability of detailed EMT models from the vendors. The order of power electronic equipment is often high, so considering the full order model will greatly increase the difficulty of analysis, especially when considering the multi-machine system. Therefore, how to balance the low order characteristics and accuracy of the analysis model and propose an effective reduced-order model for synchronous stability analysis is a problem to be studied in the future [57,121].

To fully understand the problem and implement reliable countermeasures, it is indeed necessary to develop new modeling and analysis tools that could rigorously characterize these issues. In the aspect of simulation, the electromechanical simulation model is no longer applicable. At this time, how to deal with the numerical stability problem caused by multiple time scales in the electromagnetic transient model and the simulation speed problem caused by large-scale high-order equations also need to be further studied. Emerging fields such as artificial intelligence and big data could be exploited for data-driven modeling, real-time monitoring and analysis, and adaptive control of emerging stability issues [116,125,133–135]. Ref. [136] provides a comprehensive survey of WTG modeling considering the instability type, disturbance size, and time span. However, it is challenging to develop a generic model for different converter-based equipment or CIGs because the configuration of the control structure of a CIG from different vendors could be very different and is usually kept confidential. Nevertheless, a template model of CIGs can be developed, which could be used as a generic model where a modeling component from different vendors can be easily plugged to obtain a fully functional detailed model [136].

The increasing proportion of converter-based equipment in the distribution system will also affect the system-wide stability. The new family inverters so-called grid-forming converters are also being adopted by some vendors. The level of details considered in the analytical modeling of converter-based equipment for studying different emerging stability issues is important. The modeling details of converter-based equipment depend on the stability problem being study. The fast developments are increasing the complexity of the stability phenomena, and thus increasing the need for more efficient and standardized modeling techniques for stability analysis.

One of the major limitations of the linearized modeling-based analysis techniques is that the analysis results are valid only at a certain operating point. Another limitation is that the control structure and parameters of each converter-based equipment should be known beforehand, which is quite challenging. Recently, there have been some efforts to develop accurate and efficient dynamic phasor-based interface models for time-domain hybrid simulation [21,137]. At present, commercial software and hardware simulation platforms (such as PSCAD, ETAP's eMT, and RTDS) offer multi-time step interface models such as transmission lines or interface transformers for hybrid simulations. In Refs. [138,139], a scalable and efficient multi-rate co-simulation platform is developed, which is implemented as an offline tool and an FPGA-based real-time simulator. Its effectiveness was verified by investigating actual SSR/SSO events that happened in large-scale power systems [70]. Ref. [140] proposed shifted-frequency models of power converters, a promising technique for the efficient and accurate simulation of a system with multiple MMCs. Indeed, more rigorous efforts are needed for more accurate, efficient, and fast methods for hybrid modeling and simulation analysis of dual high-penetrated power systems.

#### 6.3.2. Modeling in frequency-domain

Frequency domain impedance modeling and analysis have become very common for investigating wideband oscillation [21,141]. Some of the main advantages include 1) equipment can be modeled without complete knowledge of equipment control structure and parameters, the so-called grey/black box devices, 2) the impedance models of individual converter-based equipment and of course other system components can be combined to form a network of impedances, which would facilitate system stability analysis in the frequency-domain, 3) impedance modeling and stability analysis can be done in different time-scales to study multi-frequency instabilities. The impedance model of any power electronic equipment can be obtained using the harmonic linearization method, where a small perturbation signal of a certain frequency is injected, and the response is measured. The impedance measurement can be done on either EMT-based software simulations or real-time simulations. The impedance models can then be combined in either source-load configuration or impedance network model to apply one of the stability analysis criteria such as generalized Nyquist and reactance-frequency crossover criteria [142]. The correctness of the stability results depends on the accuracy of the device or system level impedance models. The accuracy of impedance models of a power electronic converter equipment depends on whether the details such as PLL, outer control loops, and nonlinearities are accounted for in the impedance modeling [143]. For example, for SSO analysis the outer loop control should also be considered because its bandwidth is close to or in the sub/super-synchronous frequency range. The following are some of the outstanding research prospects in this domain.

- **Considering the frequency coupling effects:** To obtain an accurate impedance model, the frequency coupling effect of the power electronic converter control should be considered [144–147]. For a grid-tied power electronic converter device, if a grid disturbance causes the voltage deviation component with frequency of $f_s$ at the connection point, the output angle of converter's PLL, the effective value of dc voltage and ac voltage will have the complement frequency component with frequency of $2f_0-f_S$, where $f_0$ is the utility or fundamental frequency. After the current internal loop control and PWM, the corresponding frequency $f_s$ and complementary frequency $2f_0-f_S$ components will be generated in the converter current. Ref. [148] established frequency coupled impedance models of converter-based WTGs and verified them through controller-hardware in the loop simulations. Ref. Developed a frequency coupled impedance model of a PV inverter [149].

The inclusion or exclusion of frequency coupling effects can impact the stability results and analysis difficulty as well. Generally, it is recommended to consider frequency coupling effects for oscillatory instabilities with lower frequencies, such as low to medium frequency ranges. For higher harmonic/oscillation frequencies, for instance, greater than four times the fundamental frequency, the frequency coupling effect is quite weak, and ignoring it would have little influence on the stability analysis results. If frequency coupling effects are considered, the impedance model would be a two-dimensional matrix, increasing the model identification and analysis difficulty.

- **Inclusion of nonlinearities:** The nonlinearities such as the PWM and current limiters in the inner control loops play an important role in defining the magnitude of the sustained wideband oscillation. Ref. [19,20,150,151] proposed a large-signal impedance method to account for the nonlinearities in the converter control. Yanhui et al. used the describing function method to model the nonlinearities and generalized Nyquist criterion to perform SSO analysis in wind farms [152]. These are some of the initial works by Tianhao et al. [153,154] on accounting for various nonlinearities in the converter control for stability analysis, however, this area requires further in-depth investigations to clarify and quantify the role of nonlinearities in wideband oscillation.
- **Operating point independency:** The initial mechanism investigations on the wideband oscillations show that the emerging stability issues are very sensitive to system operating conditions, including wind speed, solar irradiance, grid strength, and converter control structures and parameters. However, the existing impedance modeling methods are operating point specific and need to be computed again if the operating point changes. This mainly applies to nonlinear/converter-based devices, including the WTGs, PV inverters, FACTS devices, and so on. For linear components, such as transmission lines, transformers, etc., the respective impedance models can be obtained directly without considering the variations in the operating conditions. Although there have been some initial efforts to partially or completely decouple the impedance model from the operating points [155,156], this is one of the major challenges that open up a challenge to be solved by the researchers.
- **Challenges in the frequency domain analysis methods:** There are two widely used methods for impedance-based analysis 1) source-load impedance analysis, and 2) impedance network model analysis. The source-load impedance method assumes that the two partitioned subsystems are open-loop stable. Thus, the stability results get affected by the choice of the partitioning point, which divides a system into two source and load subsystems. The impedance network method assumes that all of the system modes are available at the node from where the impedance of the whole system is being viewed/aggregated. The aggregated impedance may not reflect the dynamics of some of the local modes (if there are any), which are not be observable at the aggregation node.

The stability source-load modeled system can be assessed from the phase margins. The smaller the phase margin, the more the system is vulnerable to become unstable. A specific threshold for the phase margin to indicate marginal stability for practical power systems has not been defined yet. It largely depends on the system operator's acceptance of risks and on the accuracy of the models used in determining the stability. For the impedance network-based analysis method, the system modes can be obtained by solving the zeros of the aggregated impedance. For an oscillation mode, the damping ratio and oscillatory frequency can be calculated. In an actual power system, those unstable or under-damped modes would dominate the system dynamics following a disturbance. The electrical damping of the system gives clear and meaningful information about the system's stability. To this date, no direct relationship has been established between the phase margin and electrical damping.

## 7. Concluding remarks

Modern power systems with a high proportion of renewables and power electronic converters introduce new device- and system-level features. This paper reviewed such features and their impact on the classical power system stability as well as the emerging stability challenges such as the wideband oscillation. The concluding remarks are summarized in the following points.

- The paper briefly reviewed the existing power system stability classifications proposed in 2004 and 2020 and identified their limited applicability and validity in the context of dual high-penetration power systems. It is pointed out that the extended and revisited classification of 2020 ignored the basic classification methodology adopted in 2004 that aligned the subbranches based on system variable, disturbance size, time-span, etc. Moreover, the paper critically assessed the coverage of newly added terms of 'converter driven stability' and 'resonance stability'.
- This paper proposed a new classification framework, which classified the power system stability into fundamental component and non-fundamental component stability. In the fundamental component stability branch, the concept of 'rotor angle stability' is revised to 'angle/synchronous stability', which refers to the ability of TGs as well as CIGs to maintain the synchronous operation with other equipment or the whole system. The non-fundamental component stability is further divided into three subcategories based on the frequency of the oscillation component: sub-/super-synchronous, medium/intermediate frequency, and high-frequency resonance/oscillation. The proposed classification framework maintains the inherent logic of the classical classification method and has good coverage, scalability, future adaptability, and conducive to better understanding the stability of modeling, analysis, and control.
- The paper also pointed out future classification-oriented research directions towards understanding the mechanism, modeling, analysis, and control of the various stability issues influenced or caused by the high-penetration of renewables and power electronics.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

This work is partly supported by National Natural Science Foundation of China (51737007, 51925701) and UNSW-Tsinghua Collaborative Research Seed Grants.

## References

[1] Renewable energy statistics 2020. International Renewable Energy Agency; 2020. https://irena.org/publications/2020/Jul/Renewable-energy-statistics-2020%0A. [Accessed 11 February 2021].

[2] Chakraborty A. Advancements in power electronics and drives in interface with growing renewable energy resources. Renew Sustain Energy Rev 2011;15:1816–27. https://doi.org/10.1016/j.rser.2010.12.005.

[3] Zhang G, Li Z, Zhang B, Halang WA. Power electronics converters: past, present and future. Renew Sustain Energy Rev 2018;81:2028–44. https://doi.org/10.1016/j.rser.2017.05.290.

[4] Maza-Ortega JM, Acha E, García S, Gómez-Expósito A. Overview of power electronics technology and applications in power generation transmission and distribution. J Mod Power Syst Clean Energy 2017;5:499–514. https://doi.org/10.1007/s40565-017-0308-x.

[5] Chakraborty S, Kramer B, Kroposki B. A review of power electronics interfaces for distributed energy systems towards achieving low-cost modular design. Renew Sustain Energy Rev 2009;13:2323–35. https://doi.org/10.1016/j.rser.2009.05.005.

[6] Matallana A, Ibarra E, López I, Andreu J, Garate JI, Jordà X, et al. Power module electronics in HEV/EV applications: new trends in wide-bandgap semiconductor

technologies and design aspects. Renew Sustain Energy Rev 2019;113:109264. https://doi.org/10.1016/j.rser.2019.109264.
[7] Bose BK. Power electronics in smart grid and renewable energy systems. Proc IEEE 2017;105. https://doi.org/10.1109/JPROC.2017.2752538. 2007–10.
[8] Yi W, Hill DJ, Song Y. Impact of high penetration of renewable resources on power system transient stability. IEEE Power and Energy Soc Gener Meet 2019; 2019. https://doi.org/10.1109/PESGM40551.2019.8974119. Augus:5–9.
[9] Kundur P, Paserba J, Ajjarapu V, Andersson G, Bose A, Canizares C, et al. Definition and classification of power system stability. IEEE Trans Power Syst 2004;19:1387–401. https://doi.org/10.1109/TPWRS.2004.825981.
[10] Hatziargyriou N, Milanović J, Rahmann C, Ajjarapu V, Cañizares C, Erlich I, et al. Stability definitions and characterization of dynamic behavior in systems with high penetration of power electronic interfaced technologies. 2020.
[11] Beza M, Bongiorno M. Impact of converter control strategy on low- and high-frequency resonance interactions in power-electronic dominated systems. Int J Electr Power Energy Syst 2020;120:105978. https://doi.org/10.1016/j.ijepes.2020.105978.
[12] Yuan X, Hu J, Cheng S. Multi-time scale dynamics in power electronics-dominated power systems. Front Mech Eng 2017;12:303–11. https://doi.org/10.1007/s11465-017-0428-z.
[13] Li Y, Fan L, Miao Z. Wind in weak grids: low-frequency oscillations, subsynchronous oscillations, and torsional interactions. IEEE Trans Power Syst 2019;8950. https://doi.org/10.1109/tpwrs.2019.2924412. 1–1.
[14] Ratnam KS, Palanisamy K, Yang G. Future low-inertia power systems: requirements, issues, and solutions - a review. Renew Sustain Energy Rev 2020; 124:109773. https://doi.org/10.1016/j.rser.2020.109773.
[15] Fernández-Guillamón A, Gómez-Lázaro E, Muljadi E, Molina-García Á. Power systems with high renewable energy sources: a review of inertia and frequency control strategies over time. Renew Sustain Energy Rev 2019;115:109369. https://doi.org/10.1016/j.rser.2019.109369.
[16] Fernández-Guillamón A, Gómez-Lázaro E, Muljadi E, Molina-García Á. Power systems with high renewable energy sources: a review of inertia and frequency control strategies over time. Renew Sustain Energy Rev 2019;115:109369. https://doi.org/10.1016/j.rser.2019.109369.
[17] Elnaggar AK, Rueda JL, Erlich I. Comparison of short-circuit current contribution of Doubly-Fed induction generator based wind turbines and synchronous generator. In: 2013 IEEE grenoble conference PowerTech. vol. 2013; 2013. https://doi.org/10.1109/PTC.2013.6652307. POWERTECH.
[18] Murad MAA, Ortega A, Milano F. Impact on power system dynamics of PI control limiters of VSC-Based devices. In: 20th power systems computation conference, PSCC. 2018; 2018. https://doi.org/10.23919/PSCC.2018.8442549.
[19] Wu T, Xie X, Jiang Q, Shen Z. Impedance modelling of grid-connected voltage-source converters considering the saturation non-linearity. IET Gener, Transm Distrib 2020;14:4815–23. https://doi.org/10.1049/iet-gtd.2020.0919.
[20] Shah S, Koralewicz P, Gevorgian V, Wallen R, Jha K, Mashtare D, et al. Large-signal impedance-based modeling and mitigation of resonance of converter-grid systems. IEEE Trans Sustain Energy 2019;10:1439–49. https://doi.org/10.1109/TSTE.2019.2903478.
[21] Shair J, Xie X, Liu W, Li X, Li H. Modeling and stability analysis methods for investigating subsynchronous control interaction in large-scale wind power systems. Renew Sustain Energy Rev 2021;135:110420. https://doi.org/10.1016/j.rser.2020.110420.
[22] Qin X, Zeng P, Zhou Q, Dai Q, Chen J. Study on the development and reliability of HVDC transmission systems in China. In: 2016 IEEE international conference on power system technology. vol. 2016. POWERCON; 2016. p. 1–6. https://doi.org/10.1109/POWERCON.2016.7753862.
[23] Haileselassie TM, Uhlen K. Power system security in a meshed north sea HVDC grid. Proc IEEE 2013;101:978–90. https://doi.org/10.1109/JPROC.2013.2241375.
[24] Impram S, Varbak Nese S, Oral B. Challenges of renewable energy penetration on power system flexibility: a survey. Energy Strat Rev 2020;31:100539. https://doi.org/10.1016/j.esr.2020.100539.
[25] Huang SH, Schmall J, Conto J, Adams J, Zhang Y, Carter C. Voltage control challenges on weak grids with high penetration of wind generation: ERCOT experience. IEEE Power and Energy Soc Gener Meet 2012:1–7. https://doi.org/10.1109/PESGM.2012.6344713.
[26] Wang L, Xie X, Jiang Q, Liu H, Li Y, Liu H. Investigation of SSR in practical DFIG-based wind farms connected to a series-compensated power system. IEEE Trans Power Syst 2015;30:2772–9. https://doi.org/10.1109/TPWRS.2014.2365197.
[27] Liu H, Xie X, He J, Xu T, Yu Z, Zhang C. Subsynchronous interaction between direct-drive PMSG based wind farms and weak AC networks. IEEE Trans Power Syst 2017;32:4708–20. https://doi.org/10.1109/tpwrs.2017.2682197.
[28] Buchhagen C, Rauscher C, Menze A, Jung J. BorWin1 - first experiences with harmonic interactions in converter dominated grids. In: International ETG congress 2015. Die Energiewende - Blueprints for the New Energy Age; 2015. p. 27–33.
[29] Saad H, Fillion Y, Deschanvres S, Vernay Y, Dennetiere S. On resonances and harmonics in HVDC-MMC station connected to AC grid. IEEE Trans Power Deliv 2017;32:1565–73. https://doi.org/10.1109/TPWRD.2017.2648887.
[30] Zou C, Rao H, Xu S, Li Y, Li W, Chen J, et al. Analysis of resonance between a VSC-HVDC converter and the AC grid. IEEE Trans Power Electron 2018;33:10157–68. https://doi.org/10.1109/TPEL.2018.2809705.
[31] Man J, Xie X, Xu S, Zou C, Yin C. Frequency-coupling impedance model based analysis of a high-frequency resonance incident in an actual MMC-HVDC system. IEEE Trans Power Deliv 2020;8977. https://doi.org/10.1109/tpwrd.2020.3022504. 1–1.
[32] Flynn D, Rather Z, Ardal A, D'Arco S, Hansen AD, Cutululis NA, et al. Technical impacts of high penetration levels of wind power on power system stability. Wiley Interdiscipl Rev: Energy Environ 2017;6:1–19. https://doi.org/10.1002/wene.216.
[33] Xiao X, Luo C, Zhang J, Yang W, Wu Y. Analysis of frequently over-threshold subsynchronous oscillation and its suppression by subsynchronous oscillation dynamic suppressor. IET Gener, Transm Distrib 2016;10:2127–37. https://doi.org/10.1049/iet-gtd.2015.1153.
[34] Tamimi B, Canizares C, Bhattacharya K. System stability impact of large-scale and distributed solar photovoltaic generation: the case of Ontario, Canada. IEEE Trans Sustain Energy 2013. https://doi.org/10.1109/TSTE.2012.2235151.
[35] Knüppel T, Nielsen JN, Jensen KH, Dixon A, Ostergaard J. Small-signal stability of wind power system with full-load converter interfaced wind turbines. IET Renew Power Gener 2012. https://doi.org/10.1049/iet-rpg.2010.0186.
[36] Munkhchuluun E, Meegahapola L, Vahidnia A. Impact on rotor angle stability with high solar-PV generation in power networks. In: 2017 IEEE PES innovative smart grid technologies conference Europe, ISGT-europe 2017 - proceedings 2017;; 2018. https://doi.org/10.1109/ISGTEurope.2017.8260229. Janua:1–6.
[37] Du W, Bi J, Wang T, Wang H. Impact of grid connection of large-scale wind farms on power system small-signal angular stability. CSEE J Power and Energy Syst 2015;1:83–9. https://doi.org/10.17775/cseejpes.2015.00023.
[38] Dahal S, Mithulananthan N, Saha TK. Assessment and enhancement of small signal stability of a renewable-energy-based electricity distribution system. IEEE Trans Sustain Energy 2012;3:407–15. https://doi.org/10.1109/TSTE.2012.2187079.
[39] Baquedano-Aguilar MD, Colome DG, Aguero E, Molina MG. Impact of increased penetration of large-scale PV generation on short-term stability of power systems. In: 2016 IEEE 36th central American and Panama convention. CONCAPAN 2016; 2016. p. 1. https://doi.org/10.1109/CONCAPAN.2016.7942390.
[40] Vittal E, O'Malley M, Keane A. Rotor angle stability with high penetrations of wind generation. IEEE Trans Power Syst 2012;27:353–62. https://doi.org/10.1109/TPWRS.2011.2161097.
[41] Edrah M, Lo KL, Anaya-Lara O. Impacts of high penetration of DFIG wind turbines on rotor angle stability of power systems. IEEE Trans Sustain Energy 2015;6: 759–66. https://doi.org/10.1109/TSTE.2015.2412176.
[42] Sajadi A, Kolacinski RM, Clark K, Loparo KA. Transient stability analysis for offshore wind power plant integration planning studies - Part II: long-term faults. IEEE Trans Ind Appl 2019;55:193–202. https://doi.org/10.1109/TIA.2018.2868540.
[43] Ullah NR, Thiringer T, Karlsson D. Voltage and transient stability support by wind farms complying with the E.ON netz grid code. IEEE Trans Power Syst 2007;22: 1647–56. https://doi.org/10.1109/TPWRS.2007.907523.
[44] Mitsugi Y, Yokoyama A. Phase angle and voltage stability assessment in multi-machine power system with massive integration of PV considering PV's FRT requirements and dynamic load characteristics. In: POWERCON 2014 - 2014 international conference on power system technology: towards green, efficient and smart power system. Proceedings; 2014. p. 1112–9. https://doi.org/10.1109/POWERCON.2014.6993977.
[45] Chompoo-Inwai C, Lee WJ, Fuangfoo P, Williams M, Liao J. System impact study for the interconnection of wind generation and utility system. Conf Record Indus Commer Power Syst Tech Conf 2004;41:27–32. https://doi.org/10.1109/icps.2004.1314977.
[46] Londero RR, Affonso CDM, Vieira JPA. Long-term voltage stability analysis of variable speed wind generators. IEEE Trans Power Syst 2015;30:439–47. https://doi.org/10.1109/TPWRS.2014.2322258.
[47] Bevrani H, Ghosh A, Ledwich G. Renewable energy sources and frequency regulation: survey and new perspectives. IET Renew Power Gener 2010;4: 438–57. https://doi.org/10.1049/iet-rpg.2009.0049.
[48] Xin H, Liu Y, Wang Z, Gan D, Yang T. A new frequency regulation strategy for photovoltaic systems without energy storage. IEEE Trans Sustain Energy 2013. https://doi.org/10.1109/TSTE.2013.2261567.
[49] Rakhshani E, Luna A, Rouzbehi K, Rodriguez P, Etxeberria-Otadui I. Effect of VSC-HVDC on load frequency control in multi-area power system. In: 2012 IEEE energy conversion congress and exposition, vol. 2012. ECCE; 2012. https://doi.org/10.1109/ECCE.2012.6342219. 4432–6.
[50] Gautam D, Vittal V. Impact of DFIG based wind turbine generators on transient and small signal stability of power systems. In: 2009 IEEE power and energy society general meeting, PES '09. vol. 24; 2009. p. 1426–34. https://doi.org/10.1109/PES.2009.5275847.
[51] Eftekharnejad S, Vittal V, Heydt GT, Keel B, Loehr J. Impact of increased penetration of photovoltaic generation on power systems. IEEE Trans Power Syst 2013;28:893–901. https://doi.org/10.1109/TPWRS.2012.2216294.
[52] Hu J, Huang Y, Wang D, Yuan H, Yuan X. Modeling of grid-connected DFIG-based wind turbines for dc-link voltage stability analysis. IEEE Trans Sustain Energy 2015;6:1325–36. https://doi.org/10.1109/TSTE.2015.2432062.
[53] Ma S, Geng H, Liu L, Yang G, Pal BC. Grid-synchronization stability improvement of large scale wind farm during severe grid fault. IEEE Trans Power Syst 2017;33: 216–26. https://doi.org/10.1109/tpwrs.2017.2700050.
[54] Li Y, Fan L, Miao Z. Stability control for wind in weak grids. IEEE Trans Sustain Energy 2018;3029:1–10. https://doi.org/10.1109/TSTE.2018.2878745.
[55] Fan L. Modeling type-4 wind in weak grids. IEEE Trans Sustain Energy 2019;10: 853–64. https://doi.org/10.1109/TSTE.2018.2849849.
[56] IEEE Committee Report. Reader's guide to subsynchronous resonance. IEEE Trans Power Syst 1992;7:150–7. https://doi.org/10.1109/59.141698.

[57] Shair J, Xie X, Wang L, Liu W, He J, Liu H. Overview of emerging subsynchronous oscillations in practical wind power systems. Renew Sustain Energy Rev 2019;99: 159–68. https://doi.org/10.1016/j.rser.2018.09.047.
[58] Gross LC. Sub-synchronous grid conditions: new event, new problem, and new solutions. Washington: Western Protective Relay Conference; 2010. Spokane.
[59] Adams J, Carter C, Huang SH. ERCOT experience with Sub-synchronous Control Interaction and proposed remediation. In: Proceedings of the IEEE power engineering society transmission and distribution conference. Orlando, FL, USA: IEEE; 2012. p. 1–5. https://doi.org/10.1109/TDC.2012.6281678.
[60] Mulawarman A, Mysore P. Detection of undamped sub-synchronous oscillations of wind generators with series compensated lines. Minnesota Power Systems Conference; 2011.
[61] Irwin GD, Jindal AK, Isaacs AL. Sub-Synchronous control interactions between Type 3 wind turbines and series compensated AC transmission systems. Detroit, MI, USA, USA: IEEE Power and Energy Society General Meeting; 2011. p. 1–6. IEEE.
[62] Li Y, Fan L, Miao Z. Replicating real-world wind farm SSR events. IEEE Trans Power Deliv 2019. https://doi.org/10.1109/TPWRD.2019.2931838. 1–1.
[63] Xie X, Zhang X, Liu H, Liu H, Li Y, Zhang C. Characteristic analysis of subsynchronous resonance in practical wind farms connected to series-compensated transmissions. IEEE Trans Energy Convers 2017;32:1117–26. https://doi.org/10.1109/TEC.2017.2676024.
[64] Varma RK, Moharana A. SSR in double-cage induction generator-based wind farm connected to series-compensated transmission line. IEEE Trans Power Syst 2013; 28:2573–83. https://doi.org/10.1109/TPWRS.2013.2246841.
[65] Mohammadpour HA, Santi E, Ghaderi A. Analysis of sub-synchronous resonance in doubly-fed induction generator-based wind farms interfaced with gate – controlled series capacitor. IET Gener, Transm Distrib 2014;8. https://doi.org/10.1049/iet-gtd.2013.0643. 1998–2011.
[66] Leon AE, Solsona JA. Sub-synchronous interaction damping control for DFIG wind turbines. IEEE Trans Power Syst 2015;30:419–28. https://doi.org/10.1109/TPWRS.2014.2327197.
[67] Xu Y, Zhang M, Fan L, Miao Z. Small-signal stability analysis of type-4 wind in series compensated networks. IEEE Trans Energy Convers 2019;8969. https://doi.org/10.1109/tec.2019.2943578. 1–1.
[68] Li Y, Xia Y, Peng Y. Stability analysis of the PV generator based on describing function method. In: 2019 IEEE energy conversion congress and exposition. ECCE; 2019 2019. https://doi.org/10.1109/ECCE.2019.8912852. 476–82.
[69] El Aroudi A. Prediction of subharmonic oscillation in a PV-fed quadratic boost converter with nonlinear inductors. Proc IEEE Int Symp Circ Syst 2018;2018-May. https://doi.org/10.1109/ISCAS.2018.8351379.
[70] Shu D, Xie X, Rao H, Gao X, Jiang Q, Huang Y. Sub-and super-synchronous interactions between STATCOMs and weak AC/DC transmissions with series compensations. IEEE Trans Power Electron 2018;33:7424–37. https://doi.org/10.1109/TPEL.2017.2769702.
[71] Technical report on the events of 9 august 2019. 2019.
[72] Abdalrahman A, Isabegovic E. DolWin1 - challenges of connecting offshore wind farms. In: 2016 IEEE international energy conference. ENERGYCON; 2016 2016. https://doi.org/10.1109/ENERGYCON.2016.7513981.
[73] Cheah-Mane M, Sainz L, Liang J, Jenkins N, Ugalde-Loo CE. Criterion for the electrical resonance stability of offshore wind power plants connected through HVDC links. IEEE Trans Power Syst 2017;32:4579–89. https://doi.org/10.1109/TPWRS.2017.2663111.
[74] Zhang Y, Zhu S, Sparks R, Green I. Impacts of solar PV generators on power system stability and voltage performance. IEEE Power and Energy Soc Gener Meet 2012. https://doi.org/10.1109/PESGM.2012.6344990. 1–7.
[75] Ebrahimzadeh E, Blaabjerg F, Wang X, Bak CL. Harmonic stability and resonance analysis in large PMSG-based wind power plants. IEEE Trans Sustain Energy 2018;9:12–23. https://doi.org/10.1109/TSTE.2017.2712098.
[76] Luo A, Xie N, Shuai Z, Chen Y, Jin G, Ma F, et al. Large-scale photovoltaic plant harmonic transmission model and analysis on resonance characteristics. IET Power Electron 2015;8:565–73. https://doi.org/10.1049/iet-pel.2014.0308.
[77] Subramanian SB, Varma RK, Vanderheide T. Impact of grid voltage feed-forward filters on coupling between DC-link voltage and AC voltage controllers in smart PV solar systems. IEEE Trans Sustain Energy 2020;11:381–90. https://doi.org/10.1109/TSTE.2019.2891571.
[78] Hanif M, Khadkikar V, Xiao W, Kirtley JL. Two degrees of freedom active damping technique for LCL filter-based grid connected PV systems. IEEE Trans Ind Electron 2014;61:2795–803. https://doi.org/10.1109/TIE.2013.2274416.
[79] Xie B, Zhou L, Zheng C, Zhang Q. Stability and resonance analysis and improved design of N-paralleled grid-connected PV inverters coupled due to grid impedance. Conf Proceed - IEEE Appl Power Electron Conf Expos - APEC 2018. https://doi.org/10.1109/APEC.2018.8341036. 2018-March:362–7.
[80] Amin M, Rygg A, Molinas M. Self-synchronization of wind farm in an MMC-based HVDC system: a stability investigation. IEEE Trans Energy Convers 2017;32: 458–70. https://doi.org/10.1109/TEC.2017.2661540.
[81] Lü J, Dong P, Shi G, Cai X, Li X. Subsynchronous oscillation and its mitigation of MMC-based HVDC with large doubly-fed induction generator-based wind farm integration. Zhongguo Dianji Gongcheng Xuebao/Proceed Chin Soc Electr Eng 2015;35. https://doi.org/10.13334/j.0258-8013.pcsee.2015.19.002.
[82] Amin M, Molinas M. Understanding the origin of oscillatory phenomena observed between wind farms and HVdc systems. IEEE J Emerg Selected Top Power Electron 2017;5:378–92. https://doi.org/10.1109/JESTPE.2016.2620378.
[83] Zhang S, Jiang S, Lu X, Ge B, Peng FZ. Resonance issues and damping techniques for grid-connected inverters with long transmission cable. IEEE Trans Power Electron 2014;29:110–20. https://doi.org/10.1109/TPEL.2013.2253127.
[84] Bradt M, Badrzadeh B, Camm E, Mueller D, Schoene J, Siebert T, et al. Harmonics and resonance issues in wind power plants. Proceed IEEE Power Eng Soc Trans Distribut Conf 2012. https://doi.org/10.1109/TDC.2012.6281633. 1–8.
[85] Shuai Z, Liu D, Shen J, Tu C, Cheng Y, Luo A. Series and parallel resonance problem of wideband frequency harmonic and its elimination strategy. IEEE Trans Power Electron 2014;29:1941–52. https://doi.org/10.1109/TPEL.2013.2264840.
[86] Miao Z. Impedance-model-based SSR analysis for type 3 wind generator and series-compensated network. IEEE Trans Energy Convers 2012;27:984–91. https://doi.org/10.1109/TEC.2012.2211019.
[87] Hu J, Hu Q, Wang B, Tang H, Chi Y. Small signal instability of PLL-synchronized type-4 wind turbines connected to high-impedance AC grid during LVRT. IEEE Trans Energy Convers 2016;31:1676–87. https://doi.org/10.1109/TEC.2016.2577606.
[88] Hu J, Wang B, Wang W, Tang H, Chi Y, Hu Q. Small signal dynamics of DFIG-based wind turbines during riding through symmetrical faults in weak AC grid. IEEE Trans Energy Convers 2017;32:720–30. https://doi.org/10.1109/TEC.2017.2655540.
[89] Göksu Ö, Teodorescu R, Bak CL, Iov F, Kjær PC. Instability of wind turbine converters during current injection to low voltage grid faults and PLL frequency based stability solution. IEEE Trans Power Syst 2014;29:1683–91. https://doi.org/10.1109/TPWRS.2013.2295261.
[90] Pico HNV, Johnson BB. Transient stability assessment of multi-machine multi-converter power systems. IEEE Trans Power Syst 2019;34:3504–14. https://doi.org/10.1109/TPWRS.2019.2898182.
[91] Hu R, Hu W, Chen Z. Review of power system stability with high wind power penetration. In: Iecon 2015 - 41st annual conference of the IEEE industrial electronics society; 2015. p. 3539–44. https://doi.org/10.1109/IECON.2015.7392649.
[92] Li Y, Guo J, Zhang X, Wang S, Ma S, Zhao B, et al. Over-voltage suppression methods for the MMC-VSC-HVDC wind farm integration system. IEEE Trans Circ Syst II: Exp Briefs 2020;67:355–9. https://doi.org/10.1109/TCSII.2019.2911652.
[93] Chang Q, Ge Q, Zhang B. A novel filter design approach for mitigating motor terminal overvoltage in long cable PWM drives. In: Proceedings of the 2014 9th IEEE conference on industrial electronics and applications. ICIEA; 2014 2014. https://doi.org/10.1109/ICIEA.2014.6931459. 1798–803.
[94] Benallal MN, Ailam EH, Moussa MA, Mahieddine A. Overvoltages caused by the PWM inverter in the stator coils of asynchronous motor. In: 9th international: 2014 electric power quality and supply reliability conference; 2014. https://doi.org/10.1109/PQ.2014.6866820. PQ 2014 - Proceedings 243–6.
[95] Ghoddami H, Yazdani A. A mitigation strategy for temporary overvoltages caused by grid-connected photovoltaic systems. IEEE Trans Energy Convers 2015;30: 413–20. https://doi.org/10.1109/TEC.2014.2364033.
[96] Tao H, Hu H, Zhu X, Lei K, He Z. A multifrequency model of electric locomotive for high-frequency instability assessment. IEEE Trans Trans Electr 2020;6: 241–56. https://doi.org/10.1109/TTE.2019.2960886.
[97] He J, Vrankovic Z, Ozimek PE, Winterhalter C. Investigation of common mode current related dc-bus overvoltage in multiple converter systems. Conf Proceed - IEEE Appl Power Electron Conf Expos - APEC 2016;2016–. https://doi.org/10.1109/APEC.2016.7468005. May:1084–9.
[98] Evans RD, Crary SB, Hayward AP, Koontz JA, Kroneberg AA, Scholz HJ, et al. First report of power system stability, vol. 56; 1937. https://doi.org/10.1109/T-AIEE.1937.5057521.
[99] CIGRE. Eport on the work of study committee 13: system stability and voltage, power and frequency control-appendix I. 1958.
[100] CIGRE. Definitions of general terms relating to the stability of interconnected synchronous machines. 1966.
[101] CIGRE. Tentative classification and terminologies relating to stability problems of power systems. 1978.
[102] Task force on terms & definitions system dynamic performance subcommittee power system engineering committee. Proposed terms & definitions for power system stability. IEEE Trans Power Apparatus Syst 1982. PAS-101:1894–8.
[103] Kundur P, Paserba J, Vitet S. Overview on definition and classification of power system stability. In: CIGRE/IEEE PES international symposium quality and security of electric power delivery systems; 2003. p. 1–4. https://doi.org/10.1109/QSEPDS.2003.159786. CIGRE/PES 2003.
[104] Hatziargyriou N, Milanović JV, Rahmann C, Ajjarapu V, Cañizares C. Stability definitions and characterization of dynamic behavior in systems with high penetration of power electronic interfaced technologies. Erlich† IEEE Power & Energy Soc 2020;11:2108–17.
[105] Hatziargyriou N, Milanovic JV, Rahmann C, Ajjarapu V, Canizares C, Erlich I, et al. Definition and classification of power system stability revisited & extended. IEEE Trans Power Syst 2020. https://doi.org/10.1109/TPWRS.2020.3041774.
[106] Ballance JW, Goldberg Saul. Subsynchronous resonance in series compensated systems. Colloq Torsional Oscillat Motor Gener 1973. PAS-92:1.
[107] Farmer RG, Schwalb AL, Katz E. Navajo project report on subsynchronous resonance analysis and solutions. IEEE Trans Power Apparatus Syst 1977;96: 1226–32. https://doi.org/10.1109/T-PAS.1977.32445.
[108] Eremia M, Shahidehpour M. Handbook of electrical power system dynamics: modeling, stability, and control. 2013. https://doi.org/10.1002/9781118516072.
[109] Kundur PS. Power system stability and control. 2017. https://doi.org/10.4324/b12113. New York.
[110] Xiong L, Liu X, Wang F, Zhuo F. Static Synchronous Generator model for investigating dynamic behaviors and stability issues of grid-tied inverters. In: Conference proceedings - IEEE applied power electronics conference and

exposition - APEC, 2016–; 2016. https://doi.org/10.1109/APEC.2016.7468251. May:2742–7.
[111] Lasseter RH, Chen Z, Pattabiraman D. Grid-forming inverters: a critical asset for the power grid. IEEE J Emerg Selected Top Power Electron 2020;8:925–35. https://doi.org/10.1109/JESTPE.2019.2959271.
[112] Yang C, Huang L, Xin H, Ju P. Placing grid-forming converters to enhance small signal stability of PLL-integrated power systems. IEEE Trans Power Syst 2020; 8950:1–11. https://doi.org/10.1109/TPWRS.2020.3042741.
[113] Huang L, Xin H, Dong W, Dörfler F. Impacts of grid structure on PLL-synchronization stability of converter-integrated power systems. 2019. ArXiv.
[114] Nawaz A, Wang H, Wu Q, Kumar Ochani M. TSO and DSO with large-scale distributed energy resources: a security constrained unit commitment coordinated solution. Int Trans Electr Energy Syst 2020;30:1–26. https://doi.org/10.1002/2050-7038.12233.
[115] Yang Y, Wu L. Machine learning approaches to the unit commitment problem: current trends, emerging challenges, and new strategies. Electr J 2021;34: 106889. https://doi.org/10.1016/j.tej.2020.106889.
[116] Liu H, Qi Y, Zhao J, Bi T. Data-driven subsynchronous oscillation identification using field synchrophasor measurements. IEEE Trans Power Deliv 2021;8977: 1–11. https://doi.org/10.1109/TPWRD.2021.3054889.
[117] Luo J, Teng F, Bu S. Stability-constrained power system scheduling: a review. IEEE Access 2020;8. https://doi.org/10.1109/ACCESS.2020.3042658.
[118] Wang H, Zhang G, Hu W, Cao D, Li J, Xu S. Artificial intelligence based approach to improve the frequency control in hybrid power system. Energy Rep 2020:14–8. https://doi.org/10.1016/j.egyr.2020.11.097. 00.
[119] Hosseini MM, Parvania M. Artificial intelligence for resilience enhancement of power distribution systems. Electr J 2021;34:106880. https://doi.org/10.1016/j.tej.2020.106880.
[120] Shair J, Xie X, Yan G. Mitigating subsynchronous control interaction in wind power systems: existing techniques and open challenges. Renew Sustain Energy Rev 2019;108:330–46. https://doi.org/10.1016/j.rser.2019.04.003.
[121] Meegahapola L, Sguarezi A, Bryant JS, Gu M, Conde DER, Cunha RBA. Power system stability with power-electronic converter interfaced renewable power generation: present issues and future trends. Energies 2020;13. https://doi.org/10.3390/en13133441.
[122] Xu W, Huang Z, Xie X, Li C. Synchronized waveforms – a frontier of data-based power system and apparatus monitoring, protection and control. IEEE Trans Power Deliv 2021.
[123] Meegahapola LG, Bu S, Wadduwage DP, Chung CY, Yu X. Review on oscillatory stability in power grids with renewable energy sources: monitoring, analysis, and control using synchrophasor technology. IEEE Trans Ind Electron 2021;68: 519–31. https://doi.org/10.1109/TIE.2020.2965455.
[124] Shair J, Huang W, Xie X, Wu C, Cheng M. An extended kalman filtering based time-varying fundamental and subsynchronous frequency tracker. In: 8th renewable power generation conference (RPG 2019). Shanghai, China: IET; 2019. p. 1–8. https://doi.org/10.1049/cp.2019.0386.
[125] Shair J, Xie X, Yuan L, Wang Y, Luo Y. Monitoring of subsynchronous oscillation in a series-compensated wind power system using an adaptive extended Kalman filter. IET Renew Power Gener 2021. https://doi.org/10.1049/iet-rpg.2020.0280.
[126] Azman SK, Isbeih YJ, Moursi MS El, Elbassioni K. A unified online deep learning prediction model for small signal and transient stability. IEEE Trans Power Syst 2020;35:4585–98. https://doi.org/10.1109/TPWRS.2020.2999102.
[127] Gao B, Torquato R, Xu W, Freitas W. Waveform-based method for fast and accurate identification of subsynchronous resonance events. IEEE Trans Power Syst 2019;34:3626–36. https://doi.org/10.1109/TPWRS.2019.2904914.
[128] Xiong W, Wang L, Wu R, Qi Y, Liu H, Bi T. Decision tree based subsynchronous oscillation detection and alarm method using phasor measurement units. In: iSPEC 2020 - proceedings: IEEE sustainable power and energy conference. Energy Transition and Energy Internet; 2020. p. 2448–53. https://doi.org/10.1109/iSPEC50848.2020.9351274.
[129] Xie X, Zhan Y, Liu H, Li W, Wu C. Wide-area monitoring and early-warning of subsynchronous oscillation in power systems with high-penetration of renewables. Int J Electr Power Energy Syst 2019;108:31–9. https://doi.org/10.1016/j.ijepes.2018.12.036.
[130] Xie X, Zhan Y, Shair J, Ka Z, Chang X. Identifying the source of subsynchronous control interaction via wide-area monitoring of sub/super-synchronous power flows. IEEE Trans Power Deliv 2020. https://doi.org/10.1109/TPWRD.2019.2963336. 1–1.
[131] Xie X, Liu W, Liu H, Du Y, Li Y. A system-wide protection against unstable SSCI in series-compensated wind power systems. IEEE Trans Power Deliv 2018;33: 3095–104. https://doi.org/10.1109/TPWRD.2018.2829846.
[132] Xie X, Zhan Y, Liu H, Liu C. Improved synchrophasor measurement to capture sub/super-synchronous dynamics in power systems with renewable generation. IET Renew Power Gener 2019;13:49–56. https://doi.org/10.1049/iet-rpg.2018.5533.
[133] Shair J, Xie X, Yang J, Li J, Li H. Adaptive damping control of subsynchronous oscillation in DFIG-based wind farms connected to series-compensated network. IEEE Trans Power Deliv 2021.
[134] Ma Y, Huang Q, Zhang Z, Cai D. Application of multisynchrosqueezing transform for subsynchronous oscillation detection using PMU data. IEEE Trans Ind Appl 2021. https://doi.org/10.1109/tia.2021.3057313. 9994:1–1.
[135] Damas RN, Son Y, Yoon M, Kim SY, Choi S. Subsynchronous oscillation and advanced analysis: a review. IEEE Access 2020;8:224020–32. https://doi.org/10.1109/ACCESS.2020.3044634.
[136] He X, Geng H, Mu G. Modeling of wind turbine generators for power system stability studies: a review. Renew Sustain Energy Rev 2021;143. https://doi.org/10.1016/j.rser.2021.110865.
[137] Mahseredjian J, Dinavahi V, Martinez JA. Simulation tools for electromagnetic transients in power systems: overview and challenges. IEEE Trans Power Deliv 2009;24:1657–69. https://doi.org/10.1109/TPWRD.2008.2008480.
[138] Shu D, Xie X, Zhang S, Jiang Q. Hybrid method for numerical oscillation suppression based on rational-fraction approximations to exponential functions. IET Gener, Transm Distrib 2016;10:2825–32. https://doi.org/10.1049/iet-gtd.2016.0489.
[139] Shu D, Xie X, Jiang Q, Guo G, Wang K. A multirate EMT Co-simulation of large AC and MMC-based MTDC systems. IEEE Trans Power Syst 2018;33:1252–63. https://doi.org/10.1109/TPWRS.2017.2734690.
[140] Shu D, Dinavahi V, Xie X, Jiang Q. Shifted frequency modeling of hybrid modular multilevel converters for simulation of MTDC grid. IEEE Trans Power Deliv 2018; 33:1288–98. https://doi.org/10.1109/TPWRD.2017.2749427.
[141] Shah S, Koralewicz P, Gevorgian V, Liu H, Fu J. Impedance methods for analyzing stability impacts of inverter-based resources: stability analysis tools for modern power systems. IEEE Electr Mag 2021;9:53–65. https://doi.org/10.1109/MELE.2020.3047166.
[142] Xiong L, Liu X, Liu Y, Zhuo F. Modeling and stability issues of voltage-source converter dominated power systems: a review. CSEE J Power and Energy Syst 2020:1–18. https://doi.org/10.17775/CSEEJPES.2020.03590.
[143] Liu H, Xie X, Member S, Liu Z. Comparative studies on the impedance models of VSC-based renewable generators for SSI stability analysis. IEEE Trans Energy Convers 2019:1. https://doi.org/10.1109/TEC.2019.2913778.
[144] Liu W, Xie X, Shair J, Liu H, He J. Frequency-coupled impedance model-based sub-synchronous interaction analysis for direct-drive wind turbines connected to a weak AC grid. IET Renew Power Gener 2019;13:2966–76. https://doi.org/10.1049/iet-rpg.2019.0326.
[145] Liu W, Xie X, Shair J, He J. Frequency-coupled impedance model based subsynchronous oscillation analysis for direct-drive wind turbines connected to a weak AC power system. J Eng 2019;2019:4841–6. https://doi.org/10.1049/joe.2018.9297.
[146] Liu W, Lu Z, Wang X, Xie X. Frequency-coupled admittance modelling of grid-connected voltage source converters for the stability evaluation of subsynchronous interaction. IET Renew Power Gener 2018;13:285–95. https://doi.org/10.1049/iet-rpg.2018.5462.
[147] Liu W, Xie X, Liu H, He J. Mechanism and characteristic analyses of subsynchronous oscillations caused by the interactions between direct-drive wind turbines and weak AC power systems. J Eng 2017;2017:1651–6. https://doi.org/10.1049/joe.2017.0612.
[148] Liu W, Xie X, Zhang X, Li X. Frequency-coupling admittance modeling of converter-based wind turbine generators and the control-hardware-in-the-loop validation. IEEE Trans Energy Convers 2020;35:425–33. https://doi.org/10.1109/TEC.2019.2940076.
[149] Song S, Wei Z, Lin Y, Liu B, Liu H, Member S. Impedance modeling and stability analysis of PV grid-connected inverter system considering frequency coupling. CSEE J Power and Energy Syst 2020;6:279–90. https://doi.org/10.17775/cseejpes.2019.02430.
[150] Shah S, Parsa L. Large-signal impedance for the analysis of sustained resonance in grid-connected converters. In: 2017 IEEE 18th workshop on control and modeling for power electronics (COMPEL). IEEE; 2017. p. 1–8. https://doi.org/10.1109/COMPEL.2017.8013411.
[151] Shah S, Koralewicz P, Gevorgian V, Wallen R. Large-signal impedance modeling of three-phase voltage source converters. IECON 2018 - 44th Ann Conf IEEE Indus Electron Soc 2018;15:4033–8. https://doi.org/10.1109/IECON.2018.8592852. IEEE.
[152] Xu Y, Gu Z, Sun K. Characterization of subsynchronous oscillation with wind farms using describing function and generalized nyquist criterion. IEEE Trans Power Syst 2020;35:2783–93. https://doi.org/10.1109/TPWRS.2019.2962506.
[153] Wu T, Jiang Q, Shair J, Mao H, Xie X. Inclusion of current limiter nonlinearity in the characteristic analysis of sustained subsynchronous oscillations in grid-connected PMSGs. IEEE Trans Energy Convers 2020;8969. https://doi.org/10.1109/TEC.2020.3045296. 1–1.
[154] Wu T, Luo Y, Jiang Q, Mao H, Shair J, Xie X, et al. Quantitative analysis of sustained oscillation associated with saturation non-linearity in a grid-connected voltage source converter. 865–76, https://doi.org/10.1049/rpg2.12075; 2021.
[155] Liu W, Xie X, Shair J, Li X. A nearly decoupled admittance model for grid-tied VSCs under variable operating conditions. IEEE Trans Power Electron 2020;35: 9380–9. https://doi.org/10.1109/tpel.2020.2971970.
[156] Zhang M, Wang X, Yang D, Christensen MG. Artificial neural network based identification of multi-operating-point impedance model. IEEE Trans Power Electron 2021;36:1231–5. https://doi.org/10.1109/TPEL.2020.3012136.